EECE 635: Theoretical Foundations of Security and Privacy
Fall 2026: Lectures given by Ibrahim Issa
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Chapter 1

Introduction

1.1 Privacy

Definition 1.1. Given a database D and released data set Y, if an adversary knew something about an individual i in the database, then the adversary should not be able to learn anything new about i from Y. In reality this definition is too ambitious! For example, if in a cancer research paper we learn that people who smoke are more prone to cancer, then if we know that a person i smokes, we can learn something new about i from the paper.

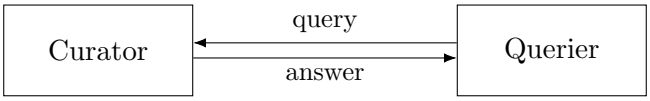


Figure 1.1: Interactive query–response model between the curator and the querier.

Remark 1.1. Aggregation also fails! For example, if we want to know what the salary of professor Ibrahim Issa is, and all we can do is ask for the average salary, then all we need to do is take the average salary from before he entered the department and after he entered the department, and we can compute his salary.

1.2 Security

Definition 1.2. Image we have a transmitter and a reciever in the perfect case there is no adversary (but if there wasn't we wouldn't need this course). If we send a message X through a channel without any encryption the adversary can intercept the message and read it without any problem. In order to prevent this we can encrypt this message using a key k and sending it through the channel using an encryption function $Y = Enc(X, K)$ the reciever then decrypts the message using a decryption function $X = Dec(Y, K)$. In the perfect case $X \perp Y$ meaning that they are independent, this is called the Shannon cipher system which achieves perfect secrecy. The problem with this is that the key must be as long as the message which is highly impractical. This is where the fields diverge into two

(1) Cryptography: where we assume that the adversary has limited computational power and we can use this to our advantage to design encryption schemes that are secure against polynomial time adversaries.

(2) Information-theoretic security: where we relax dependency using information theoretic measures.

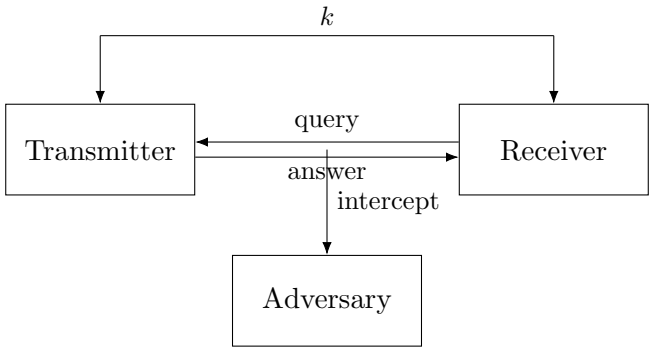


Figure 1.2: Secure communication model with a shared key k between transmitter and receiver, subject to interception by an adversary.

1.3 Warner's Randomized Response

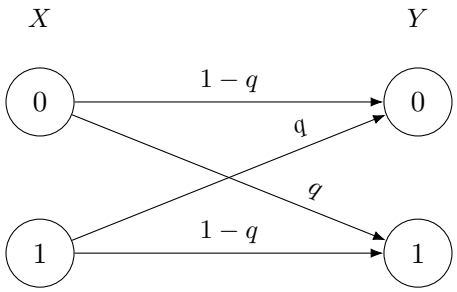


Figure 1.3: Binary channel with crossover probability q, mapping $X \in \{0, 1\}$ to $Y \in \{0, 1\}$.

Consider the above binary symmetric channel. We can use this to describe a randomized response mechanism. Let X be the true answer to a sensitive question, and let Y be the reported answer. The mechanism works as follows:

(1) Flip a biased coin with probability $1 - q$ of heads.

(2) If the coin comes up heads, report the true answer X.

(3) If the coin comes up tails, report the opposite answer $1 - X$.

So consider the case in which I ask a sensitive question like "Have you ever cheated on an exam?" and let the true answer be Bernoulli with parameter θ:

$$X \sim \text{Bernoulli}(\theta), \qquad P(X = 1) = \theta, \quad P(X = 0) = 1 - \theta.$$

The reported answer Y is also Bernoulli, with parameter

$$P(Y = 1) = (1 - q)\theta + q(1 - \theta)$$

For independent reported answers $Y_1, \ldots, Y_n$, we have

$$P_{Y_1,\ldots,Y_n}(y_1 \ldots y_n) = \prod_{i=1}^{n} P_{Y_i}(y_i) = P_Y(1)^{n_1} P_Y(0)^{n-n_1}$$

where n_1 is the number of reported answers that are 1. Now we use log likelihood to estimate θ:

$$\begin{aligned}
\ell(\theta) &= n_1 \log P_Y(1) + (n - n_1) \log P_Y(0) \\
&= n_1 \log((1-q)\theta + q(1-\theta)) + (n-n_1)\log((1-q)(1-\theta) + q\theta) \\
&= n_1 \log((1-2q)\theta + q) + (n-n_1)\log((1-2q)(1-\theta) + q),
\end{aligned}$$

Now taking the derivative with respect to θ gives us:

$$\frac{\partial \ell}{\partial \theta} = \frac{n_1(1-2q)}{(1-2q)\theta + q} - \frac{(n-n_1)(1-2q)}{(1-2q)(1-\theta)+q}.$$

Setting this derivative to zero and solving for θ gives us the maximum likelihood estimate:

$$\hat{\theta} = \frac{n_1/n - q}{1-2q}$$

Now lets check our estimate is unbiased:

$$\begin{aligned}
\mathbb{E}[\hat{\theta}] &= E\left[\frac{n_1/n - q}{1-2q}\right] \\
&= \frac{E[n_1/n] - q}{1-2q} \\
&= \frac{P_Y(1) - q}{1-2q} \\
&= \frac{(1-q)\theta + q(1-\theta) - q}{1-2q} \\
&= \frac{(1-q)\theta - q\theta}{1-2q} \\
&= \frac{(1-2q)\theta}{1-2q} \\
&= \theta
\end{aligned}$$

Now lets compute the variance of our estimate:

$$\begin{aligned}
\mathrm{Var}(\hat{\theta}) &= \frac{P_Y(1)(1-P_Y(1))}{n(1-2q)^2} \\
&= \frac{\big(q + (1-2q)\theta\big)\big(1 - q - (1-2q)\theta\big)}{n(1-2q)^2} \\
&= \frac{1}{n}\left((\theta - \theta^2) + \frac{q(1-q)}{(1-2q)^2}\right).
\end{aligned}$$

Chapter 2

Probability Review

Definition 2.1. A probability space is a triple $(\Omega, \mathcal{E}, P)$ where:

(1) Ω is the sample space, which is the set of all possible outcomes of a random experiment.

(2) $\mathcal{E}$ is a σ-algebra of events, which is a collection of subsets of Ω satisfying the following properties:

   (1) $\emptyset \in \mathcal{E}$

   (2) If $A \in \mathcal{E}$, then $A^c \in \mathcal{E}$

   (3) If $A_1, A_2, A_3, \ldots \in \mathcal{E}$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{E}$

(3) $P : \mathcal{E} \to \mathbb{R}$ is a mapping such that:

   (1) $P(A) \geq 0$ for all $A \in \mathcal{E}$

   (2) $P(\Omega) = 1$

   (3) If $A_1, A_2, A_3, \ldots \in \mathcal{E}$ are disjoint, then $P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$

Proposition 2.1. *Some properties of probability spaces:*

(1) If $A_1, A_2, \ldots, A_n \in \mathcal{E}$ then $A_1 \cap A_2 \cap \ldots \cap A_n \in \mathcal{E}$

Proof. This follows from the fact that $\mathcal{E}$ is closed under complements and countable unions. Specifically, we can write the intersection as a complement of a union of complements:

$$A_1 \cap A_2 \cap \ldots \cap A_n = (A_1^c \cup A_2^c \cup \ldots \cup A_n^c)^c$$

Since $A_i \in \mathcal{E}$, we have $A_i^c \in \mathcal{E}$ for all i. The union of these complements is also in $\mathcal{E}$, and taking the complement again gives us an element of $\mathcal{E}$. Therefore, the intersection is in $\mathcal{E}$. □

(2) $P(\emptyset) = 0$

Proof. $P(\emptyset) = P(\emptyset \cup \emptyset) = P(\emptyset) + P(\emptyset)$, which implies $P(\emptyset) = 0$. □

(3) $P(A^c) = 1 - P(A)$ for all $A \in \mathcal{E}$

Proof. Since A and A^c are disjoint and their union is Ω, we have:

$$P(A \cup A^c) = P(\Omega) = 1$$

By the additivity property, we also have:

$$P(A \cup A^c) = P(A) + P(A^c)$$

Therefore, we can equate the two expressions:

$$1 = P(A) + P(A^c)$$

Rearranging gives us:

$$P(A^c) = 1 - P(A)$$

□

(4) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ *for all* $A, B \in \mathcal{E}$

Proof. The sets A and $B \setminus A$ are disjoint, and $A \cup B = A \cup (B \setminus A)$. Thus, by additivity,

$$P(A \cup B) = P(A) + P(B \setminus A).$$

Moreover, $A \cap B$ and $B \setminus A$ are disjoint and their union is B. Therefore,

$$P(B) = P(A \cap B) + P(B \setminus A),$$

so $P(B \setminus A) = P(B) - P(A \cap B)$. Substituting this into the first equality gives

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

□

(5) Union Bound: $P(A \cup B) \leq P(A) + P(B)$ *for all* $A, B \in \mathcal{E}$

Proof. This follows directly from the previous property, since $P(A \cap B) \geq 0$. □

2.1 Conditional Probability

Definition 2.2. Given an event B with $P(B) > 0$, the conditional probability of an event A given B is defined as:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Definition 2.3. We say that two events A and B are independent if:

$$P(A \cap B) = P(A)P(B)$$

More generally, we say that a collection of events $\{A_i\}_{i \in I}$ is independent if for every finite subset $J \subseteq I$, we have:

$$P\left(\bigcap_{j \in J} A_j\right) = \prod_{j \in J} P(A_j)$$

Theorem 2.1 (Bayes' Theorem)**.** *Let A and B be two events with $P(B) > 0$. Then:*

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} = \frac{P(A)P(B|A)}{P(A)P(B|A) + P(A^c)P(B|A^c)}$$

2.1.1 Probability Mass Function

Definition 2.4. The probability mass function (pmf) of a discrete random variable X is a map

$$p_X : \mathbb{R} \to [0, 1]$$

with:

$$p_X(x) = P(X = x)$$

for all $x \in X(\Omega)$.

Example 2.1. Consider two fair dice rolls and let X be the value of the first die, and Y be the sum of the two values. Then

$$p_Y(5) = \frac{4}{36}$$

since there are four ways to get a sum of 5: $(1, 4), (2, 3), (3, 2), (4, 1)$ and there are 36 possible outcomes in total, and

$$\mathbb{E}[X] = \frac{1}{6}\sum_{x=1}^{6} x = \frac{7}{2}$$

and

$$\mathbb{E}[Y] = \sum_{y=2}^{12} y \cdot p_Y(y) = 7$$

2.2 Discrete Random Variables

Definition 2.5. A discrete random variable X is a map $X : \Omega \to \mathbb{R}$ where:

(1) $X(\Omega)$ is finite or countably infinite

(2) For all $x \in \mathbb{R}$, $\{\omega \in \Omega : X(\omega) = x\} \in \mathcal{E}$, i.e., the preimage of x under X is an event in the σ-algebra $\mathcal{E}$ and $P(X = x) = P(\{\omega \in \Omega : X(\omega) = x\})$

The expected value of a discrete random variable X is defined as:

$$\mathbb{E}[X] = \sum_{x \in X(\Omega)} xP(X = x)$$

2.2.1 Joint Probability Distribution of Two Random Variables X and Y

Definition 2.6. Let X and Y be two discrete random variables defined on the same probability space $(\Omega, \mathcal{E}, P)$. The joint probability distribution of X and Y is defined as:

$$\begin{aligned} P(X = x, Y = y) &= P(\{\omega \in \Omega : X(\omega) = x, Y(\omega) = y\}) \\ &= P(\{\omega \in \Omega : X(\omega) = x\} \cap \{\omega \in \Omega : Y(\omega) = y\}) \end{aligned}$$

for all $x \in X(\Omega)$ and $y \in Y(\Omega)$.

Marginal Probability Distribution of X and Y

Definition 2.7. The marginal probability distribution of X is defined as:

$$P(X = x) = \sum_{y \in Y(\Omega)} P(X = x, Y = y)$$

for all $x \in X(\Omega)$. Similarly, the marginal probability distribution of Y is defined as:

$$P(Y = y) = \sum_{x \in X(\Omega)} P(X = x, Y = y)$$

for all $y \in Y(\Omega)$.

2.2.2 Functions of random variables

Definition 2.8. Let $X : \Omega \to \mathbb{R}$ be a random variable and $f : \mathbb{R} \to \mathbb{R}$ then $f(X)$ is also a random variable then

$$\mathbb{E}[f(X)] = \sum_x p(x) \cdot f(x)$$

Proposition 2.2 (Linearity of Expectation)**.** *Given a pair of real random variables X and Y and constants $a, b \in \mathbb{R}$, we have:*

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$$

Proof. Let X and Y be discrete random variables defined on the same probability space $(\Omega, \mathcal{E}, P)$. Then we can write:

$$\begin{aligned}
\mathbb{E}[aX + bY] &= \sum_{x,y} (ax + by)p(x, y) \\
&= \sum_{x,y} axp(x, y) + \sum_{x,y} byp(x, y) \\
&= a \sum_{x,y} xp(x, y) + b \sum_{x,y} yp(x, y) \\
&= a \sum_x x \sum_y p(x, y) + b \sum_y y \sum_x p(x, y) \\
&= a \sum_x xp(x) + b \sum_y yp(y) \\
&= a\mathbb{E}[X] + b\mathbb{E}[Y]
\end{aligned}$$

□

2.2.3 Independence

Definition 2.9. Two random variables X and Y are said to be independent if and only if their joint probability distribution is the product of their marginal probability distributions, i.e.,

$$P_{XY}(x, y) = P_X(x)P_Y(y)$$

for all $x \in X(\Omega)$ and $y \in Y(\Omega)$. We denote this by $X \perp Y$.

Definition 2.10. We say that a collection of random variables $\{X_i\}_{i\in I}$ are mutually independent if for all $x_1, \dots x_n$ we have

$$P_{X_i^n}(x_1, \dots, x_n) = \prod_{i=1}^{n} P_{X_i}(x_i)$$

Example 2.2. Let $X \sim Ber(\frac{1}{2})$ and $Y \sim Ber(\frac{1}{2})$ be two independent Bernoulli random variables and $Z = X \oplus Y \to Z \sim Ber(\frac{1}{2})$. Is $X \perp Z$?

$$P_{Z|X}(0|0) = P_{Y|X}(0|0) = P_Y(0) = \frac{1}{2}$$

and

$$P_{Z|X}(1|0) = \frac{1}{2}$$

$$P_{Z|X}(0|1) = P_{Y|X}(1|1) = \frac{1}{2} = P_Z(0)$$

$$P_{Z|X}(1|1) = \frac{1}{2} = P_Z(1)$$

So, $Z \perp X$.

Proposition 2.3. *If X and Y are independent then the expectation of their product is the product of their expectations, i.e.,*

$$\mathbb{E}[XY] = \mathbb{E}[X] \cdot \mathbb{E}[Y]$$

Markov Chains (Conditional Independence)

Definition 2.11. We say that $X - Y - Z$ form a markov chain if for all x, y, z where $p(x, y) > 0$ we have

$$P_{Z|X,Y}(z|x, y) = P_{Z|Y}(z|y)$$

which is equivalent to

$$P_{X|Y,Z}(x|y, z) = P_{X|Y}(x|y)$$

Equivalently,

$$P_{XYZ}(x, y, z) = P_X(x) \cdot P_{Y|X}(y|x) \cdot P_{Z|Y}(z|y)$$

Example 2.3. Given any two random variables X and Y and some deterministic function f it is always true that

$$f(X) - X - Y$$

holds

2.3 Weak Law of Large Numbers

Theorem 2.2. *Let $X_1, X_2, \dots, X_n$ be a sequence of pairwise independent random variables with common mean μ and varience σ^2, then for all $\varepsilon > 0$ we have*

$$P\left(\left|\frac{1}{n}\sum_{i=1}^{n} X_i - \mu\right| \geq \varepsilon\right) \leq \frac{\sigma^2}{n\varepsilon^2}$$

2.3.1 Nice Inequalities

Theorem 2.3 (Markov's Inequality)**.** *Given a non-negative random variable X and $a > 0$ we have*

$$P(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$$

Proof. $\mathbb{E}[X] = \sum_x p(x) \cdot x = \sum_{x<a} p(x) \cdot x + \sum_{x \geq a} p(x) \cdot x \geq 0 + a \sum_{x \geq a} p(x) = aP(X \geq a)$ □

Theorem 2.4 (Chebyshev's Inequality)**.** *Given a random variable X with mean μ and variance σ^2 then for all $\alpha > 0$ we have*

$$P(|X - \mu| > \alpha \cdot \sigma) \leq \frac{1}{\alpha^2}$$

Proof. Apply Markov's Inequality to the random variable $(X - \mu)^2$ with $a = \alpha^2\sigma^2$ so

$$\begin{aligned} P(|X - \mu| \geq \alpha \cdot \sigma) &= P((X - \mu)^2 \geq \alpha^2\sigma^2) \\ &\leq \frac{\mathbb{E}[(X - \mu)^2]}{\alpha^2\sigma^2} \\ &= \frac{1}{\alpha^2} \end{aligned}$$

□

Proof of the Weak Law of Large Numbers. Let $Y = \frac{1}{n}\sum_{i=1}^n X_i$ then $\mathbb{E}[Y] = \mu$ and $\mathrm{Var}(Y) = \frac{\sigma^2}{n}$ so by Chebyshev's Inequality we have

$$P\left(|Y - \mu| \geq \frac{\varepsilon}{\sigma_Y} \cdot \sigma_Y\right) \leq \frac{\sigma_Y^2}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2}$$

□

Remark 2.1. We used here that $\mathrm{Var}\left(\sum X_i\right) = \sum \mathrm{Var}(X_i)$ wif the X_i are pairwise independent

2.4 Hypothesis Testing

Definition 2.12. Given two distributions Q_0 and Q_1 on a common alphabet X, we define the total variation distance between them as

$$d_{TV}(Q_0, Q_1) = \sup_A \{Q_0(A) - Q_1(A)\}$$

Theorem 2.5. $d_{TV}(Q_0, Q_1) = \frac{1}{2}\sum_x |Q_0(x) - Q_1(x)|$

Proof. For any A we have $Q_0(A) - Q_1(A) = \sum_{x \in A} (Q_0(x) - Q_1(x))$. Let $A^* = \{x : Q_0(x) \geq Q_1(x)\}$. Then

Claim. For all A $Q_0(A) - Q_1(A) \leq Q_0(A^*) - Q_1(A^*)$

proof of claim.

$$\begin{aligned} \sum_{x \in A} (Q_0(x) - Q_1(x)) &= \sum_{x \in A \cap A^*} (Q_0(x) - Q_1(x)) + \sum_{x \in A \cap (A^*)^c} (Q_0(x) - Q_1(x)) \\ &\leq \sum_{x \in A \cap A^*} (Q_0(x) - Q_1(x)) + \sum_{x \in A^* \cap A^c} (Q_0(x) - Q_1(x)) \\ &= \sum_{x \in A^*} (Q_0(x) - Q_1(x)) \,. \end{aligned}$$

□

Now

$$\sum_{x \in X} (Q_0(x) - Q_1(x)) = 0$$

so

$$\sum_{Q_0 > Q_1} (Q_0(x) - Q_1(x)) + \sum_{Q_0 = Q_1} (Q_0(x) - Q_1(x)) + \sum_{Q_0 < Q_1} (Q_0 - Q_1) = 0$$

so

$$d_{TV} = \sum_{Q_0 > Q_1} (Q_0 - Q_1) = \sum_{Q_0 < Q_1} (Q_1 - Q_0)$$

Therefore,

$$\sum_x |Q_0(x) - Q_1(x)| = \sum_{Q_0 > Q_1} (Q_0 - Q_1) + \sum_{Q_0 < Q_1} (Q_1 - Q_0) = 2d_{TV}$$

□

2.4.1 Binary Hypothesis Testing

Definition 2.13. Given two distribution Q_0 and Q_1 and observe some sample X we want to decide whether $H_0 : X \sim Q_0$ (X belongs to Q_0) or $H_1 : X \sim Q_1$ (X belongs to Q_1).

Bayesian Setup

Definition 2.14. In the Bayesian setup we assume that H_1 is true with probability p and H_0 is true with probability $1 - p$. We want to minimize the probability of error. Start with a uniform prior $p = \frac{1}{2}$. We need a decision rule $g : X \to \{H_0, H_1\}$ and let H be the true distribution so we want

$$\min_g P\left(H \neq g(X)\right)$$

Can allow randomized function $G : X \to \Delta_{0,1}$ where $\Delta_{\{0,1\}}$ is the set of all distributions over $\{0, 1\}$. Our goal is to find g^* i.e the optimal rule in which

$$g^*(x) = \begin{cases} \mathrm{H}_0, & \text{if } Q_0(x) \geq Q_1(x), \\ \mathrm{H}_1, & \text{if } Q_0(x) < Q_1(x). \end{cases}$$

So let's expand

$$\begin{aligned}
P(H \neq g(X)) &= P\left(g(x) \neq 0 \mid H = H_0\right) P(H_0) + P\left(g(x) \neq 1 \mid H = H_1\right) P(H_1) \\
&= \sum_x P(X = x \mid H = H_0)\, P(g = 1 \mid X = x)\, P(H_0) \\
&\quad + \sum_x P(X = x \mid H = H_1)\, P(g = 0 \mid X = x)\, P(H_1) \\
&= \sum_x Q_0(x)\, P(g = 1 \mid X = x)\, P(H_0) + \sum_x Q_1(x)\, P(g = 0 \mid X = x)\, P(H_1) \\
&= \frac{1}{2} \sum_x Q_0(x)\, g(1 \mid x) + \frac{1}{2} \sum_x Q_1(x)\, g(0 \mid x) \\
&= \frac{1}{2} \left(\sum_x Q_0(x)\, g(1 \mid x) + \sum_x Q_1(x)\, (1 - g(1 \mid x)) \right) \\
&= \frac{1}{2} \left(1 + \sum_x (Q_0(x) - Q_1(x))\, g(1 \mid x) \right)
\end{aligned}$$

so

$$g^*(1|x) = \begin{cases} 0 & \text{if } Q_0(x) \geq Q_1(x) \\ 1 & \text{if } Q_0(x) < Q_1(x) \end{cases}$$

and

$$P_e^* = \frac{1}{2}\left(1 - \sum_{x:Q_0(x)<Q_1(x)} (Q_1(x) - Q_0(x))\right) = \frac{1}{2}\left(1 - d_{TV}(Q_0, Q_1)\right)$$

Definition 2.15. In the case of a non-uniform prior p we have

$$P(g(x) \neq H) = \sum_x Q_0(x)g(1|x)(1-p) + \sum_x Q_1(x)(1-g(1|x))p$$

and we want to minimize this. So

$$\begin{aligned} P(g(x) \neq H) &= \sum_x Q_0(x)g(1|x)(1-p) + \sum_x Q_1(x)(1-g(1|x))p \\ &= p + \sum_x g(1|x) \cdot (Q_0(x)(1-p) - Q_1(x)p) \end{aligned}$$

Therefore,

$$g^*(x) = \begin{cases} 0 & \text{if } Q_0(x)(1-p) \geq Q_1(x)p \\ 1 & \text{if } Q_0(x)(1-p) < Q_1(x)p \end{cases}$$

This is called the MAP rule (Maximum A Posteriori rule).

Remark 2.2.

$$g_{ML}(x) = \begin{cases} 0 & \text{if } Q_0(x) \geq Q_1(x) \\ 1 & \text{if } Q_0(x) < Q_1(x) \end{cases}$$

is called the maximum likelihood rule. This is the same as the MAP rule with a uniform prior.

Definition 2.16 (Neyman-Pearson)**.** In the case of no prior, we use the same setting i.e

$$H_0 : X \sim Q_0$$

and

$$H_1 : X \sim Q_1$$

We want to find a decision rule $g : X \to \{0, 1\}$. Define

$$P_{MD} = P(g(X) = 0|H_1) = Q_1(A_g^c)$$

and

$$P_{FA} = P(g(X) = 1|H_0) = Q_0(A_g)$$

Where $A_g = \{x \mid g(x) = 1\}$. Note that the points $(0, 1)$ (always miss detection) and $(1, 0)$(always false alarm) are trivially achievable.

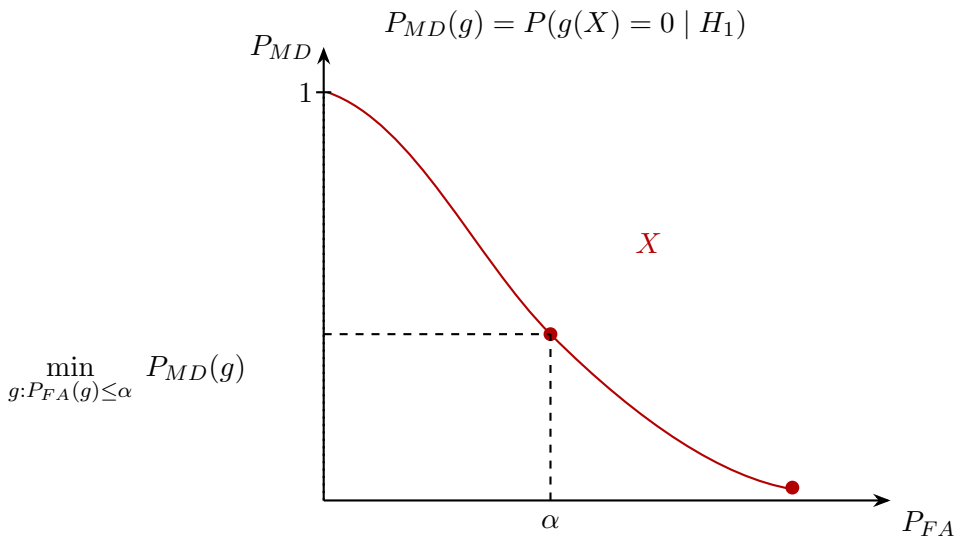


Figure 2.1: False-alarm and missed-detection trade-off.

Lemma 2.1 (Neyman-Pearson Lemma)**.** *For a given threshold $T > 0$, define*

$$g_T(x) = \begin{cases} 1 & \text{if } Q_1(x) > TQ_0(x) \\ 0 & \text{otherwise} \end{cases}$$

then for any $g : X \to \{0, 1\}$, if $P_{FA}(g) \leq P_{FA}(g_T)$ then $P_{MD}(g) \geq P_{MD}(g_T)$

Proof. Let $A_T = \{x \mid g_T(x) = 1\}$ and $B = \{x \mid g(x) = 1\}$, and recall the indicator function

$$\phi_{A_T}(x) = \begin{cases} 1 & x \in A_T \\ 0 & x \notin A_T \end{cases}$$

We claim that for all x we have

$$(\phi_{A_T}(x) - \phi_B(x)) \cdot (Q_1(x) - TQ_0(x)) \geq 0$$

Proof of Claim. We consider the claim in cases:

(1) If $x \in A_T \cap B$ or $x \notin A_T \cup B$, then $\phi_{A_T}(x) - \phi_B(x) = 0$, so the product is 0.

(2) If $x \in A_T \setminus B$ then $\phi_{A_T}(x) - \phi_B(x) = 1$. Since $x \in A_T$, we have $Q_1(x) > TQ_0(x)$, so the product is positive.

(3) If $x \in B \setminus A_T$ then $\phi_{A_T}(x) - \phi_B(x) = -1$. Since $x \notin A_T$, we have $Q_1(x) \leq TQ_0(x)$, so the product is non-negative.

□

Now we sum over all x:

$$\begin{aligned}
&\sum_{x\in X} \left(\phi_{A_T}(x) - \phi_B(x)\right)\left(Q_1(x) - TQ_0(x)\right) \geq 0 \\
&\implies \sum_{x\in X} \phi_{A_T}(x)Q_1(x) - T\sum_{x\in X} \phi_{A_T}(x)Q_0(x) - \sum_{x\in X} \phi_B(x)Q_1(x) + T\sum_{x\in X} \phi_B(x)Q_0(x) \geq 0 \\
&\implies Q_1(A_T) - TQ_0(A_T) - Q_1(B) + TQ_0(B) \geq 0 \\
&\implies 1 - P_{MD}(g_T) - TP_{FA}(g_T) - (1 - P_{MD}(g)) + TP_{FA}(g) \geq 0 \\
&\implies P_{MD}(g) - P_{MD}(g_T) \geq T\left(P_{FA}(g_T) - P_{FA}(g)\right)
\end{aligned}$$

□

2.4.2 M-ary Hypothesis Testing

Definition 2.17. Given a random variable $V \in \mathcal{V}$ be some random variable with $|\mathcal{V}| = k$. Given conditional distribution (Channel) $P(X|V)$ with X finite. Sometimes it is useful to think of $P_{X|V}$ as a matrix given by

$$P_{X|V} = \begin{bmatrix} P(X = x_1|V = v_1) & P(X = x_2|V = v_1) & \cdots & P(X = x_\ell|V = v_1) \\ P(X = x_1|V = v_2) & P(X = x_2|V = v_2) & \cdots & P(X = x_\ell|V = v_2) \\ \vdots & \vdots & \ddots & \vdots \\ P(X = x_1|V = v_k) & P(X = x_2|V = v_k) & \cdots & P(X = x_\ell|V = v_k) \end{bmatrix}$$

where $X = \{x_1, x_2, \ldots, x_\ell\}$. We want to observe X and guess V, i.e we want to find a decision rule

$$g : X \to \mathcal{V} \quad \text{such that} \quad \min_g P(g(X) \neq V)$$

. Think of someone throwing a die and telling you to guess the outcome, then any guess is a good one; While if one throws a biased die then the good guess is the most biased one. Thus, consider the function

$$g_\emptyset : \emptyset \to \mathcal{V}$$

given by

$$g_\emptyset = \text{argmax}_v\, P_V(v)$$

but argmax may not be unique so we change this by just taking an element i.e

$$g_\emptyset \in \text{argmax}_v\, P_V(v)$$

Therefore $P(g_\emptyset = V) = \max_v P_V(v)$. Now lets actually find our decision rule,

$$g : X \to \mathcal{V}$$

Then,

$$g_{MAP}(x) \in \text{argmax}_v\, P(v \mid x)$$

Why? Since

$$P(g(X) \neq V) = \sum_x P_X(x)P(g(x) \neq V|X = x)$$

and we want to minimize this, so we want to minimize $P(g(x) \neq V|X = x)$ for all x. But

$$P(g(x) \neq V|X = x) = 1 - P(g(x) = V|X = x)$$

so we want to maximize $P(g(x) = V|X = x)$ for all x. But

$$P(g(x) = V|X = x) = P(V = g(x)|X = x) = p(v = g(x)|x)$$

so we want to maximize $p(v|x)$ over all v for each x. Then

$$Pr(g_{MAP}(X) = V) = \sum_x P_X(x) \max_v P(v|x) = \sum_x \max_v P_{VX}(v, x) = \sum_x \max_v P_{X|V}(x|v)P_V(v)$$

Chapter 3

Privacy

3.1 Privacy Setups

Definition 3.1. There are two main privacy setups:

(1) Trusted curator: There is a trusted curator who has access to the data and can release some information about it. The curator is trusted to not leak any information about the data.

(2) Non-trusted curator (Local privacy): There is no trusted curator and the data is held by the users. The users can release some information about their data, but they are not trusted to not leak any information about the data so they add noise to their data before releasing it.

Notation 3.1. We denote the set of domain of database entries by $\mathcal{X}$ and the database by $D \in \mathcal{X}^n$ where n is the size of the database.

Definition 3.2. A query is a function

$$f : \mathcal{X}^n \to \mathcal{Y}$$

where $\mathcal{Y}$ is the set of possible outputs of the query.

Definition 3.3. A counting query is a query that counts the number of entries in the database that satisfy some property. It is given by

$$f : \mathcal{X}^n \to \{0, 1, 2, \ldots, n\}$$

where the output is the number of entries that satisfy the property.

3.2 Reconstruction Attack

Definition 3.4. Consider the alphabet $\mathcal{X} = \{0, 1\}$ and $D \in \{0, 1\}^n$ and let the query be : Given $S \subseteq [n]$ where $[n] = \{1, 2, \ldots, n\}$, return $A(S) = \sum_{i \in S} D_i$. Then, if we have access to $A(S)$ for all $S \subseteq [n]$, we can reconstruct the database D by querying $A(\{i\})$ for all $i \in [n]$ and setting $D_i = A(\{i\})$. This is called a reconstruction attack. So how do we answer, let the answering mechanism be denoted by $r(S)$ where

$$r : 2^{[n]} \to \mathbb{R}$$

and define the error of the answering mechanism on a specific set to be

$$E(S, r) = |r(S) - A(S)|$$

and my answering mechanism may be randomized.

Definition 3.5. The Hamming distance given $u, v \in \mathcal{X}^n$ is defined by

$$d_H(u,v) = \sum_{i=1}^{n} \mathbb{1}_{u_i \neq v_i}$$

where $\mathbb{1}_{u_i \neq v_i}$ is the indicator function that is 1 if $u_i \neq v_i$ and 0 otherwise.

Theorem 3.1. *Suppose r satisfies that there exists E_0 such that $\forall S$ we have $E(S,r) \leq E_0$. Then, there exists an adversary that can reconstruct D within $4E_0$ positions i.e there exists m and a sequence $s_1, \ldots, s_m$ and a function $g : \mathbb{R}^m \to \mathcal{X}^n$ such that*

$$d_H\left(D, g(r(s_1), \ldots r(s_m))\right) \leq 4E_0$$

where d_H is the Hamming distance.

Example 3.1. Let $n = 5$ pick a database $D = (0,1,1,0,1)$ and let $S = \{1,3\}$ then $A(S) = 1$

Proof. Let $m = 2^n - 1$ and consider all possible nonempty subsets of $[n]$. Now we have

$$r(S_1), r(S_2), \ldots, r(S_m),$$

and we construct g as an algorithm:

$d \in \{0,1\}^n$
for $i = 1$ to m **do**
  if $|A_d(S_i) - r(S_i)| > E_0$ **then**
    reject d
accept d.

Let d^* be the accepted string and we need to show that $d_H(d^*, D) \leq 4E_0$. Let

$$I_0 = \{i \mid D_i = 0\}$$

and

$$I_1 = \{i \mid D_i = 1\}$$

Consider

$$\begin{aligned} |A_{d^*}(I_0) - A_D(I_0)| &= |A_{d^*}(I_0) - r(I_0) + r(I_0) - A_D(I_0)| \\ &\leq |A_{d^*}(I_0) - r(I_0)| + |r(I_0) - A_D(I_0)| \\ &\leq E_0 + E_0 \\ &= 2E_0. \end{aligned}$$

Similarly, for I_1 we have

$$|A_{d^*}(I_1) - A_D(I_1)| \leq 2E_0$$

Now,

$$d_H(d^*, D) = \sum_{i=1}^{n} \mathbb{1}_{d_i^* \neq D_i} = |A_{d^*}(I_0) - A_D(I_0)| + |A_{d^*}(I_1) - A_D(I_1)| \leq 2E_0 + 2E_0 = 4E_0$$

as desired. □

Consequence 3.1. Say $E_0 = \frac{n}{400}$ i.e $d_H(d^*, D) \leq \frac{n}{100}$ which means that d^* is correct to 99% of the positions. This means that if we want to answer all queries with error at most $\frac{n}{400}$ then we cannot guarantee any privacy.

3.3 Differential Privacy

Definition 3.6. A possibly randomized algorithm M also called a mechanism is a mapping $M : \mathcal{X}^n \to \mathcal{Y}$

Definition 3.7. Given $\varepsilon \geq 0$ a mechanism $M : \mathcal{X}^n \to \mathcal{Y}$ is ε-DP (ε-Differential Privacy) if for all events and for all neighboring databases i.e $D, D' \in \mathcal{X}^n$ such that $d_H(D, D') = 1$ denoted by $D \sim D'$ we have

$$P(M(D) \in S) \leq e^{\varepsilon} P(M(D') \in S)$$

for all $S \subseteq \mathcal{Y}$ and

$$P(M(D') \in S) \leq e^{\varepsilon} P(M(D) \in S)$$

for all $S \subseteq \mathcal{Y}$ since $D \sim D'$ is symmetric. This means that the output of the mechanism does not depend too much on any single entry in the database.

Remark 3.1. By Problem Set 1, we have that the max ratio of the probabilities of sets is equal to max ratio of the probabilities of points, i.e

$$\max_{S \subseteq X} \frac{P(S)}{Q(S)} = \max_{x \in X} \frac{P(x)}{Q(x)}$$

Where P and Q are two distributions over the same set X. Therefore, we can check the DP condition by checking the ratio of probabilities of points instead of sets.

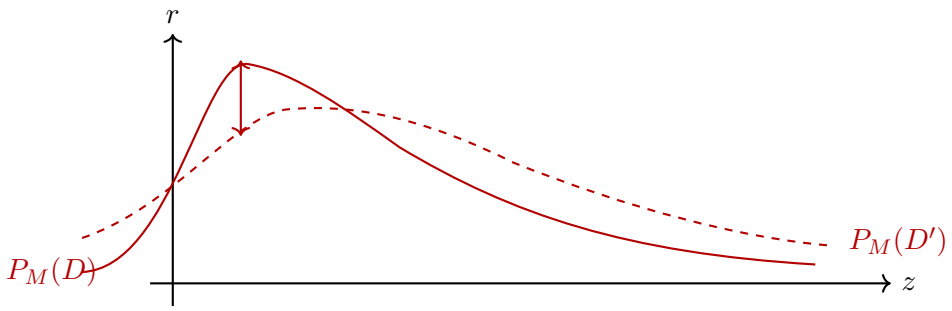


Figure 3.1: ε-DP: output distributions $P_M(D)$ and $P_M(D')$ on neighboring datasets D, D'.

Definition 3.8. In the case where $\varepsilon = 0$ we call this perfect privacy, i.e the output of the mechanism is independent of the input database. In the case where $\varepsilon = \infty$ we call this no privacy, i.e the output of the mechanism can depend arbitrarily on the input database.

Remark 3.2. ε can't be negative since

$$\max_{x} \frac{P(x)}{Q(x)} \leq \alpha < 1$$

implies that one of the distributions is not a distribution since it will always dominate the other distribution.

Proposition 3.1. *These are our assumptions:*

(1) No assumptions on the distribution of the database D.

(2) "Strong" adversary

Definition 3.9 (Approximate Differential Privacy)**.** Given $\varepsilon, \delta \geq 0$ a mechanism $M : \mathcal{X}^n \to \mathcal{Y}$ is (ε, δ)-DP if for all events and for all neighboring databases i.e $D, D' \in \mathcal{X}^n$ such that $d_H(D, D') = 1$ denoted by $D \sim D'$ we have

$$P(M(D) \in S) \leq e^{\varepsilon} P(M(D') \in S) + \delta$$

for all $S \subseteq \mathcal{Y}$ and

$$P(M(D') \in S) \leq e^{\varepsilon} P(M(D) \in S) + \delta$$

for all $S \subseteq \mathcal{Y}$ since $D \sim D'$ is symmetric. This means that the output of the mechanism does not depend too much on any single entry in the database.

Interpretation 3.1. δ is relaxing the constraints think of it as a margin of error. It is the probability that the mechanism fails to be ε-DP. In practice, we want δ to be very small, e.g 10^{-6} or smaller.

Proposition 3.2 (Post-Processing)**.** *Let $M : \mathcal{X}^n \to \mathcal{Y}$ be an (ε, δ)-differentially private mechanism, and let $f : \mathcal{Y} \to \mathcal{Z}$ be an arbitrary (possibly randomized) function. Then the composition*

$$f \circ M \ : \ \mathcal{X}^n \longrightarrow \mathcal{Z}$$

is also (ε, δ)-differentially private.

In other words, no post-processing of a differentially private output — deterministic or randomized, and performed without access to the original dataset — can weaken its privacy guarantee.

Proof. For simplicity assume that f is deterministic. We want to show that for all $D \sim D'$ and for all $T \subseteq Z$ we have

$$P(f(M(D)) \in T) \leq e^{\varepsilon} P(f(M(D')) \in T) + \delta$$

Let $S_T = \{y \in Y : f(y) \in T\}$ then $f(M(D)) \in T$ if and only if $M(D) \in S_T$. Then

$$P(f \circ M(D) \in T) = P(M(D) \in S_T)$$

which implies that

$$P(f \circ M(D) \in T) \leq e^{\varepsilon} P(M(D') \in S_T) + \delta = e^{\varepsilon} P(f \circ M(D') \in T) + \delta$$

as desired. □

3.4 Warner's Randomized Response

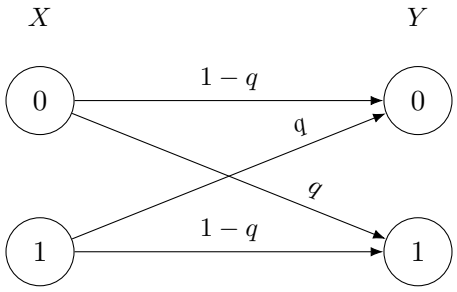


Figure 3.2: Binary channel with crossover probability q, mapping $X \in \{0, 1\}$ to $Y \in \{0, 1\}$.

Question 3.1. Is Warner's randomized response mechanism (ε)-DP? If so, what is the value of ε in terms of q?

Answer 3.1. Let $D = [0]$ and $D' = [1]$ be two neighboring databases. Then, for any $S \subseteq \{0, 1\}$ we have

- If $S = \emptyset$ or $S = \{0, 1\}$ trivial.
- If $S = \{0\}$ then
$$(1 - q) \leq e^{\varepsilon} q$$
and
$$q \leq e^{\varepsilon}(1 - q)$$
- If $S = \{1\}$ then
$$q \leq e^{\varepsilon}(1 - q)$$
and
$$(1 - q) \leq e^{\varepsilon} q$$

Therefore, Warner's randomized response mechanism is (ε)-DP with

$$\varepsilon = \log\left(\max\left\{\frac{1-q}{q}, \frac{q}{1-q}\right\}\right)$$

as desired.

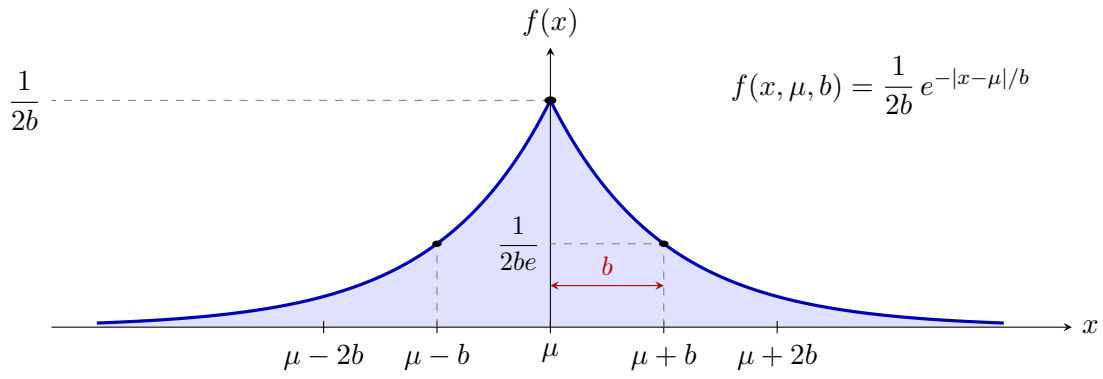
Remark 3.3. Later we will see that if M is (ε, δ)-DP then if we take $M \times M : \mathcal{X}^n \to \mathcal{Y} \times \mathcal{Y}$ is $(?, ?)$-DP.

3.5 Laplace Mechanism

Definition 3.10. Laplace distribution centered at μ with scale $b > 0$ is defined as the following pdf

$$\text{Lap}(x; b, \mu) = \frac{1}{2b} e^{-\frac{|x-\mu|}{b}}$$

with $\text{Lap}(x; b, 0) = \text{Lap}(b)$ with mean 0. So, if $X \sim \text{Lap}(b, \mu)$ we have $\mathbb{E}[X] = \mu$ and $\text{Var}(X) = 2b^2$. The Laplace distribution is symmetric about μ and has heavier tails than the normal distribution. The Laplace distribution is often used in differential privacy to add noise to data in order to protect individual privacy.



Definition 3.11 (Laplace Mechanism)**.** Given $f : \mathcal{X}^n \to \mathbb{R}$ we define

$$M_L(D, f, \varepsilon) = f(D) + Y$$

with $Y \sim \text{Lap}(\alpha)$

Example 3.2. Counting Query: Let $D = \{0,1\}^n$ and $T \subseteq [n]$ and define $f_T(D) = \sum_{i \in T} D_i$

Theorem 3.2. *The Laplace mechanism with* $\alpha = \frac{\sup_{D \sim D'} |f(D) - f(D')|}{\varepsilon}$ *is* ε*-DP.*

Proof. Fix $D \sim D'$ and $S \subseteq \mathbb{R}$. We need to verify that

$$P(M(D) \in S) \leq e^{\varepsilon} P(M(D') \in S).$$

Now

$$\begin{aligned}
\frac{P[M(D) \in S]}{P[M(D') \in S]} &= \frac{\int_S P_D(z)\, dz}{\int_S P_{D'}(z)\, dz} \\
&= \frac{\int_S \text{Lap}\big(z;\, \alpha, f(D)\big)\, dz}{\int_S \text{Lap}\big(z;\, \alpha, f(D')\big)\, dz} \\
&= \frac{\int_S \exp\big(-\frac{1}{\alpha}\,|z - f(D)|\big)\, dz}{\int_S \exp\big(-\frac{1}{\alpha}\,|z - f(D')|\big)\, dz} \\
&\leq \sup_{z \in \mathbb{R}} \frac{\exp\big(-\frac{1}{\alpha}\,|z - f(D)|\big)}{\exp\big(-\frac{1}{\alpha}\,|z - f(D')|\big)} \\
&= \sup_{z \in \mathbb{R}} \exp\Big(\tfrac{1}{\alpha}\big(|z - f(D')| - |z - f(D)|\big)\Big) \\
&\leq \exp\Big(\tfrac{1}{\alpha}\,|f(D') - f(D)|\Big).
\end{aligned}$$

We want to choose α so that

$$\exp\Big(\tfrac{1}{\alpha}\,|f(D') - f(D)|\Big) \leq e^{\varepsilon},$$

that is,

$$\frac{1}{\alpha}\,|f(D) - f(D')| \leq \varepsilon \qquad \forall\, D \sim D'.$$

Hence we may take

$$\alpha = \frac{\sup\limits_{D \sim D'} |f(D) - f(D')|}{\varepsilon}.$$

□

Definition 3.12 (Laplace Mechanism for Vector-Valued Functions)**.** Given $f : \mathcal{X}^n \to \mathbb{R}^k$ we define

$$M_L(D, f, \varepsilon) = f(D) + Y$$

with $Y \sim \text{Lap}(\alpha)^k$ i.e each component of Y is independently drawn from $\text{Lap}(\alpha)$ and

$$\alpha = \frac{\sup_{D \sim D'} \left\| f(D) - f(D') \right\|_1}{\varepsilon}$$

Example 3.3. One example of a vector-valued function is the histogram query. One cool thing is that in the histogram query we will always have

$$\Delta f = \sup_{D \sim D'} \|f(D) - f(D')\|_1 = 2$$

Definition 3.13. The infinity norm for a vector $z \in \mathbb{R}^k$ is defined as

$$\|z\|_\infty = \max_{1 \le i \le k} |z_i|$$

Theorem 3.3. *Given $f : \mathcal{X}^n \to \mathbb{R}^k$ and $\varepsilon, \delta > 0$ then the Laplace mechanism with $\alpha = \frac{\Delta f}{\varepsilon}$ satisfies:*

$$P\left[\|M_L(D, f, \varepsilon) - f(D)\|_\infty \ge \frac{\Delta f}{\varepsilon} \ln \frac{k}{\delta} \right] \le \delta$$

i.e. δ bounds the probability that the error exceeds $\frac{\Delta f}{\varepsilon} \ln \frac{k}{\delta}$.

Exercise 3.1. Show that if $Y \sim \text{Lap}(b)$ then

$$P(|Y| > tb) \le e^{-t}$$

Proof. — At home □

Proof of Theorem. We want to look at

$$\begin{aligned}
&P\left(\max_{1 \le i \le k} |M_L(D)_i - f(D)_i| \ge \frac{\Delta f}{\varepsilon} \ln \frac{k}{\delta} \right) \\
&= P\left(\max_{1 \le i \le k} |Y_i| \ge \frac{\Delta f}{\varepsilon} \ln \frac{k}{\delta} \right) \\
&= P\left(\bigcup_{i=1}^{k} \left\{ |Y_i| \ge \frac{\Delta f}{\varepsilon} \ln \frac{k}{\delta} \right\} \right) \\
&\le \sum_{i=1}^{k} P\left(|Y_i| \ge \frac{\Delta f}{\varepsilon} \ln \frac{k}{\delta} \right) \\
&\le k e^{-\ln \frac{k}{\delta}} \\
&= k \frac{\delta}{k} \\
&= \delta
\end{aligned}$$

□

3.6 Gaussian Mechanism

Definition 3.14. Given a query $f : \mathcal{X}^n \to \mathbb{R}^k$ we defined the gaussian mechanism as

$$M_G(D, f, \varepsilon) = f(D) + Y^k$$

where $Y_i \sim \mathcal{N}(0, \sigma^2)$ independent and identically distributed.

Theorem 3.4. *The gaussian mechanism is not ε-DP for any ε.*

Proof. Want that for all $D \sim D'$ with $S \subseteq \mathbb{R}$ we have

$$\frac{P(M(D) \in S)}{P(M(D') \in S)} \leq e^{\varepsilon}$$

we proceed similarly as we did in the proof of the Laplace mechanism. We have

$$\begin{aligned}
\sup_S \frac{P(M(D) \in S)}{P(M(D') \in S)} &= \sup_S \frac{\int_S \mathcal{N}(f(D), \sigma^2)dy}{\int_S \mathcal{N}(f(D'), \sigma^2)dy} \\
&= \sup_{y \in \mathbb{R}} \frac{e^{-\frac{1}{2\sigma^2}(y - f(D))^2}}{e^{-\frac{1}{2\sigma^2}(y - f(D'))^2}} \\
&= \sup_{y \in \mathbb{R}} e^{\frac{1}{2\sigma^2}\left((y - f(D'))^2 - (y - f(D))^2\right)} \\
&= \sup_{y \in \mathbb{R}} e^{\frac{1}{2\sigma^2}(f(D) - f(D'))(2y - f(D) - f(D'))} \\
&= \infty
\end{aligned}$$

Therefore the gaussian mechanism is not ε-DP for any ε. □

Theorem 3.5. *For $\varepsilon \in (0, 1)$, the gaussian mechanism $M_G(D, f, \varepsilon)$ is (ε, δ)-DP with*

$$\sigma = \frac{\Delta_{2,f}\sqrt{2\ln\left(\frac{1.25}{\delta}\right)}}{\varepsilon}$$

where $\Delta_{2,f} = \sup_{D \sim D'} \|f(D) - f(D')\|_2$ is the L_2 sensitivity of f.

Proof. — In the appendix of the book. □

3.7 Exponential Mechanism

Example 3.4. Say we have an unlimited supply of something, and we have a bunch of users who want to buy it and they place bids we will create a mechanism that is ε-DPsuch that it will take the bids and outputs a number that is close to the maximum bid but it will not reveal the maximum bid. For simplicity assume we have 3 clients willing to pay 1 dollar, and 1 client willing to pay 4 dollars. Let x denote the selling price:

- For $x \leq 1$ we get $4x$
- For $1 < x \leq 4$ we get x
- For $x > 4$ we get 0

Definition 3.15. We define the utility function or score as a map

$$u : \mathcal{X}^n \times \mathcal{Y} \to \mathbb{R}$$

Definition 3.16. The exponential mechanism is denoted as

$$M_E(D, u, \mathcal{Y})$$

such that it selects $y \in \mathcal{Y}$ with probability $\propto e^{\frac{\varepsilon}{2\Delta}u(D,y)}$ i.e

$$P(y \mid D) = \frac{e^{\frac{\varepsilon}{2\Delta}u(D,y)}}{\sum_{y'} e^{\frac{\varepsilon}{2\Delta}u(D,y')}}$$

Theorem 3.6. *$M_E(D, u, \mathcal{Y})$ is ε-DP*

Proof. Fix y and $D \sim D'$ and solve

$$\begin{aligned}
\frac{P(y|D)}{P(y|D')} &= \frac{e^{?u(D,y)}}{e^{?u(D',y)}} \cdot \frac{\sum_{y'} e^{?u(D',y')}}{\sum_{y'} e^{?u(D,y')}} \\
&= e^{?u(D,y)-?u(D',y)} \cdot \frac{\sum_{y'} e^{?u(D',y')}}{\sum_{y'} e^{?u(D,y')}} \\
&\leq e^{?\Delta} \cdot e^{?\Delta} \\
&= e^{2?\Delta}
\end{aligned}$$

with $\Delta = \sup_{D\sim D'} \max_y |u(D, y) - u(D', y)|$ so we want to solve for ? such that $2?\Delta = \varepsilon$ i.e $? = \frac{\varepsilon}{2\Delta}$ as desired. □

Remark 3.4. Smaller ε means more privacy but less utility. Larger ε means less privacy but more utility.

Definition 3.17. The optimal utility for a given database D is defined as

$$OPT_u(D) = \max_{y\in\mathcal{Y}} u(D, y)$$

this is not necessarily unique, there may be multiple y that achieve the maximum utility.

Theorem 3.7. *Let $R_{OPT}(D) = \{y \in \mathcal{Y} : u(D, y) = OPT_u(D)\}$ be the set of optimal solutions. Then, for all $\varepsilon > 0$*

$$P\left[u\left(D, M_E(D, u, \mathcal{Y})\right) \leq OPT_u(D) - \frac{2\Delta}{\varepsilon}\left(\ln\frac{|\mathcal{Y}|}{|R_{OPT}|} + t\right)\right] \leq e^{-t}$$

Proof. Fix $c > 0$ then

$$\begin{aligned}
P\left[u\left(D, M_E(D, u, \mathcal{Y})\right) \leq c\right] &= \sum_{y:u(D,y)\leq c} \frac{e^{\frac{\varepsilon}{2\Delta}u(D,y)}}{\sum_{y'} e^{\frac{\varepsilon}{2\Delta}u(D,y')}} \\
&\leq \frac{|\mathcal{Y}|e^{\frac{\varepsilon}{2\Delta}c}}{|R_{OPT}|e^{\frac{\varepsilon}{2\Delta}OPT_u(D)}} \\
&= \frac{|\mathcal{Y}|}{|R_{OPT}|}e^{\frac{\varepsilon}{2\Delta}(c-OPT_u(D))}
\end{aligned}$$

□

3.8 Composition Theorems

Definition 3.18. Let $M_i : \mathcal{X}^n \to \mathcal{Y}_\rangle$ be some mechanism such that its $(\varepsilon_i, \delta_i)$-DP for all $1 \leq i \leq k$. Define

$$M_{[k]} : \mathcal{X}^n \to \prod_{i=1}^{k} \mathcal{Y}_\rangle$$

as

$$M_{[k]}(D) = (M_1(D), \ldots, M_k(D))$$

then $M_{[k]}$ is $(\sum_{i=1}^k \varepsilon_i, \sum_{i=1}^k \delta_i)$-DP.

3.8.1 Divergences, Jensen's Inequality and Convex Functions

Definition 3.19 (Max Divergence)**.** Given two distributions P and Q on a common alphabet $\mathcal{X}$ we define the max divergence as

$$D_\infty(P\|Q) = \ln \sup_{S \subseteq \mathcal{X}} \frac{P(S)}{Q(S)}$$

Definition 3.20. (δ-approximate max divergence) Given $\delta \in (0, 1)$, let P and Q be two distributions on a common alphabet $\mathcal{X}$ we define the δ-approximate max divergence as

$$D_\infty^\delta(P\|Q) = \ln \sup_{S \subseteq \mathcal{X}} \frac{P(S) - \delta}{Q(S)}$$

Theorem 3.8. *M is (ε, δ)-DP if and only if for all neighboring databases $D \sim D'$ we have*

$$D_\infty^\delta(P_D\|P_{D'}) \leq \varepsilon$$

and

$$D_\infty^\delta(P_{D'}\|P_D) \leq \varepsilon$$

where P_D and $P_{D'}$ are the induced distributions of $M(D)$ and $M(D')$ respectively.

Definition 3.21 (KL Divergence or Relative Entropy)**.** Given two distributions P and Q on a common alphabet $\mathcal{X}$ we define the KL divergence as

$$D_{KL}(P\|Q) = \mathbb{E}_p\left(\ln \frac{P(X)}{Q(X)}\right)$$

In the discrete case we have

$$D_{KL}(P\|Q) = \sum_{x \in \mathcal{X}} P(x) \ln \frac{P(x)}{Q(x)}$$

and in the continuous case we have

$$D_{KL}(P\|Q) = \int_{\mathcal{X}} P(x) \ln \frac{P(x)}{Q(x)} dx$$

Convention 3.1. For all $p > 0$, we have

$$p \ln \frac{p}{0} = \infty$$

and for all q we have

$$0 \ln \frac{0}{q} = 0$$

.

Proposition 3.3. *$D_{KL}(P\|Q) \geq 0$ with equality if and only if $P = Q$.*

Definition 3.22. A set is convex if for all $x, y \in S$ and for all $\lambda \in [0,1]$ we have $\lambda x + (1-\lambda)y \in S$. A function $f : S \to \mathbb{R}$ is convex if for all $x, y \in S$ and for all $\lambda \in [0,1]$ we have

$$f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y)$$

Theorem 3.9. *Jensen's Inequality If f is convex and X is a random variable then*

$$f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)]$$

Remark 3.5. If f is concave then

$$f(\mathbb{E}[X]) \geq \mathbb{E}[f(X)]$$

Remark 3.6. If f is strictly convex then equality holds if and only if X is a constant random variable.

Remark 3.7. The KL divergence is not symmetric, i.e., $D_{KL}(P\|Q) \neq D_{KL}(Q\|P)$ in general.

Proof of Proposition. Recall that

$$D_{KL}(P\|Q) = \mathbb{E}_p\left(\ln \frac{P(X)}{Q(X)}\right) = \sum_x P(x) \ln \frac{P(x)}{Q(x)}.$$

We can write

$$\begin{aligned} D_{KL}(P\|Q) &= \mathbb{E}_p\left(-\ln \frac{Q(X)}{P(X)}\right) \\ &\geq -\ln \mathbb{E}_p\left(\frac{Q(X)}{P(X)}\right) \\ &= -\ln \sum_x P(x)\frac{Q(x)}{P(x)} \\ &= -\ln \sum_x Q(x) \\ &= 0 \end{aligned}$$

□

3.8.2 Adaptive Composition

Motivation 3.1. Given a family of mechanisms $\mathcal{F}$ and an adversary $\mathcal{A}$. Fix $b \in \{0,1\}$ unknown to the adversary $\mathcal{A}$ and consider the following experiment: for $i = 1, \ldots, k$ do the following:

(1) $\mathcal{A}$ outputs two neighboring databases $D^{(0,i)}, D^{(1,i)}$

(2) $\mathcal{A}$ chooses a mechanism $M_i \in \mathcal{F}$

(3) $\mathcal{A}$ receives $y_i \sim M_i(D^{(b,i)})$

Adversary $\mathcal{A}$ needs to find b. Adversary's view is $V^b = (R^b, Y_1^b, \ldots, Y_k^b)$ where R^b is the internal randomness of $\mathcal{A}$.

Definition 3.23. We say a class of mechanisms $\mathcal{F}$ satisfies (ε, δ)-DP under k-fold composition if for all adversaries $\mathcal{A}$ and all $S \subseteq R \times \mathcal{Y}_1 \times \cdots \times \mathcal{Y}_k$ we have

$$P(V^1 \in S) \leq e^{\varepsilon} P(V^0 \in S) + \delta$$

and

$$P(V^0 \in S) \leq e^{\varepsilon} P(V^1 \in S) + \delta$$

Theorem 3.10. *Let $\mathcal{F}$ be a class of ε-DP mechanisms, then for all $\delta' > 0$ F is (ε', δ')-DP under k-fold composition with*

$$\varepsilon' = \varepsilon\sqrt{2k\ln\left(\frac{1}{\delta'}\right)} + k\varepsilon(e^{\varepsilon} - 1)$$

Lemma 3.1. *Given two distributions P and Q such that*

$$D_{\infty}(P\|Q) \leq \varepsilon$$

and

$$D_{\infty}(Q\|P) \leq \varepsilon$$

Then we can bound the KL-divergence as

$$D_{KL}(P\|Q) \leq \varepsilon(e^{\varepsilon} - 1)$$

Proof.

$$\begin{aligned}
D_{KL}(P\|Q) + D_{KL}(Q\|P) &= \sum_x P(x)\ln\frac{P(x)}{Q(x)} + \sum_x Q(x)\ln\frac{Q(x)}{P(x)} \\
&= \sum_x (P(x) - Q(x))\ln\frac{P(x)}{Q(x)} \\
&\leq \sum_x |P(x) - Q(x)| \cdot |\ln\frac{P(x)}{Q(x)}| \\
&\leq \sum_x |P(x) - Q(x)| \cdot \varepsilon \\
&= \sum_x (\max(P(x), Q(x)) - \min(P(x), Q(x))) \cdot \varepsilon \\
&\leq \varepsilon \sum_x \min(P(x), Q(x))(e^{\varepsilon} - 1) \\
&= \varepsilon(e^{\varepsilon} - 1)\sum_x \min(P(x), Q(x)) \\
&\leq \varepsilon(e^{\varepsilon} - 1)
\end{aligned}$$

□

Lemma 3.2. *$D_{\infty}^{\delta}(P\|Q) \leq \varepsilon$ if and only if there exists a distribution R such that $d_{TV}(R, P) \leq \delta$ and $D_{\infty}(R\|Q) \leq \varepsilon$.*

Proof.

- ($\Longleftarrow$): We need to show that for all $S \subseteq \mathcal{X}$ we have

  $$P(S) \leq e^{\varepsilon}Q(S) + \delta$$

  Given $R(S) \leq e^{\varepsilon}Q(S)$ and $|P(S) - R(S)| \leq \delta$. So we have, $P(S) \leq R(S) + \delta \leq e^{\varepsilon}Q(S) + \delta$.

- ($\Longrightarrow$): We need to construct R. We are given that $P(S) \leq e^{\varepsilon}Q(S) + \delta$ for all $S \subseteq \mathcal{X}$. Let

  $$S = \{x : P(x) > e^{\varepsilon}Q(x)\}$$

  and

  $$T = \{x : P(x) < Q(x)\}$$

and

$$(S \cup T)^c = \{x : Q(x) \leq P(x) \leq e^{\varepsilon} Q(x)\}$$

Then define R as follows:

$$R(x) = \begin{cases} e^{\varepsilon} Q(x) & \text{if } x \in S \\ P(x) & \text{if } x \in (S \cup T)^c \\ P(x) \leq R(x) \leq Q(x) & \text{if } x \in T \end{cases}$$

So we need $Q(T) + 1 - P(S) - P(T) + e^{\varepsilon} Q(S) \geq 1$ since this is the maximum choice for $R(T)$ and we need $P(T) + 1 - P(S) - P(T) + e^{\varepsilon} Q(S) \leq 1$ (Note that the second contraint we have by construction of S and T). Rewrite the first inequality as

$$\begin{aligned} Q(T) - P(T) &= \big(1 - Q(T^c)\big) - \big(1 - P(T^c)\big) \\ &= P(T^c) - Q(T^c) \\ &= \sum_{x \in T^c} \big(P(x) - Q(x)\big) \\ &\geq \sum_{x \in S} \big(P(x) - Q(x)\big) \\ &= P(S) - Q(S) \\ &\geq P(S) - e^{\varepsilon} Q(S). \end{aligned}$$

We still need $d_{TV}(R, P) \leq \delta$ so

$$d_{TV} = P(S) - R(S) = P(S) - e^{\varepsilon} Q(S) \leq \delta$$

Since $D_{\infty}^{\delta}(P \| Q) \leq \varepsilon$ we have $\ln \frac{P(S) - \delta}{Q(S)} \leq \varepsilon$.

□

Lemma 3.3 (Azuma's Inequality)**.** *Let $C_1, C_2, \ldots, C_k$ be real valued random variables such that there exists α such that for all i we have*

$$P(|C_i| \leq \alpha) = 1$$

and there exists β such that for all i and for all $(c_1, \ldots, c_{i-1})$ in support we have

$$\mathbb{E}[C_i \mid C_1 = c_1, \ldots, C_{i-1} = c_{i-1}] \leq \beta$$

then for all $z > 0$,

$$P\left(\sum_{i=1}^{k} C_i > k\beta + z \cdot \alpha \cdot \sqrt{k}\right) \leq e^{-\frac{z^2}{2}}$$

Proof of Composition Theorem. We want to show that for all $S \subseteq R \times \mathcal{Y}_1 \times \cdots \times \mathcal{Y}_k$ we have

$$P(V^0 \in S) \leq e^{\varepsilon'} P(V^1 \in S) + \delta'$$

We begin by defining

$$B = \{v \colon P(V^0 = v) > e^{\varepsilon'} P(V^1 = v)\}$$

so its sufficient to show that $P(V^0 \in B) \leq \delta'$. Let $Z = \ln \frac{P(V^0=V)}{P(V^1=V)}$ where $V \sim V^0$. So, $V = (r, y_1, \ldots, y_k)$

$$P(V^0 = V) = P_R(r) \times P(Y_1^0 = y_1 \mid R = r) \times \cdots \times P(Y_k^0 = y_k \mid R = r, Y_1^0 = y_1, \ldots, Y_{k-1}^0 = y_{k-1})$$

Let

$$Z = \sum_{i=1}^{k} \ln \frac{P\big(Y_i^0 = y_i \mid Y_1^0 = y_1, \ldots, Y_{i-1}^0 = y_{i-1}\big)}{P\big(Y_i^1 = y_i \mid Y_1^1 = y_1, \ldots, Y_{i-1}^1 = y_{i-1}\big)},$$

and denote the i-th term by C_i. By assumption (ε-DP family) we have $|C_i| \leq \varepsilon$, and

$$\begin{aligned}
\mathbb{E}_{V^0}\big[C_i \mid Y_1^0 = y_1, \ldots, Y_{i-1}^0 = y_{i-1}\big] &= \sum_{y_i} P\big(Y_i^0 = y_i \mid Y_1^0 = y_1, \ldots, Y_{i-1}^0 = y_{i-1}\big) \\
&\qquad \cdot \ln \frac{P\big(Y_i^0 = y_i \mid Y_1^0 = y_1, \ldots, Y_{i-1}^0 = y_{i-1}\big)}{P\big(Y_i^1 = y_i \mid Y_1^1 = y_1, \ldots, Y_{i-1}^1 = y_{i-1}\big)} \\
&= D_{\mathrm{KL}}\Big(P_{Y_i^0 \mid Y_1^0 = y_1, \ldots, Y_{i-1}^0 = y_{i-1}} \\
&\qquad \Big\| P_{Y_i^1 \mid Y_1^1 = y_1, \ldots, Y_{i-1}^1 = y_{i-1}}\Big).
\end{aligned}$$

We know that

$$D_\infty(P_{Y_i^0 \mid Y_1^0 = y_1, \ldots, Y_{i-1}^0 = y_{i-1}} \,\|\, P_{Y_i^1 \mid Y_1^1 = y_1, \ldots, Y_{i-1}^1 = y_{i-1}}) \leq \varepsilon$$

and

$$D_\infty(P_{Y_i^1 \mid Y_1^1 = y_1, \ldots, Y_{i-1}^1 = y_{i-1}} \| P_{Y_i^0 \mid Y_1^0 = y_1, \ldots, Y_{i-1}^0 = y_{i-1}}) \leq \varepsilon$$

which implies that

$$\mathbb{E}_{V_0}[C_i \mid Y_1^0 = y_1, \ldots, Y_{i-1}^0 = y_{i-1}] \leq \varepsilon(e^\varepsilon - 1)$$

Finally

$$\begin{aligned}
P(Z > \varepsilon') &= P(\sum_{i=1}^{k} C_i > \varepsilon') \\
&= P(\sum_{i=1}^{k} C_i > k\varepsilon(e^\varepsilon - 1) + \sqrt{2 \ln \frac{1}{\delta'}} \cdot \varepsilon \cdot \sqrt{k}) \\
&\leq e^{-\frac{1}{2}\left(\sqrt{2 \ln \frac{1}{\delta'}}\right)^2} \\
&= \delta'
\end{aligned}$$

□

3.8.3 Above Threshold Algorithm

Algorithm 1 AboveThreshold$(D, \{f_i\}, T, \varepsilon)$

Require: Database D, adaptively chosen queries $f_1, f_2, \ldots$ of ℓ_1-sensitivity 1, threshold T, privacy parameter ε

Ensure: Stream of responses in $\{\uparrow, \downarrow\}^*$

```
T̂ ← T + Lap(2/ε)
for each query f_i do
    ν_i ← Lap(4/ε)
    if f_i(D) + ν_i > T̂ then
        output ↑ and halt
    else
        output ↓
```

Proposition 3.4. *AboveThreshold is ε-DP.*

Proof. Let $a = \downarrow, \downarrow, \ldots, \uparrow$ be a possible output of k queries. Let

$$A = AbTh(D, \{f_i\}, T, \varepsilon)$$

and

$$A' = AbTh(D', \{f_i\}, T, \varepsilon)$$

We find the probabilites

$$P(A = a) = P(\{f_i(D) + \nu_i \leq \hat{T}, i = 1, \ldots, k-1\} \bigcap \{f_k(D) + \nu_k > \hat{T}\})$$

Fix $\nu_1, \ldots, \nu_{k-1}$ and consider

$$P(A = a \mid \nu_1, \ldots, \nu_{k-1})$$

then

$$P_{\hat{T},\nu_k}(A = a \mid V_1^{k-1}) = P_{\hat{T},\nu_k}(\{f_i(D) + \nu_i \leq \hat{T}, i = 1, \ldots, k-1\} \bigcap \{f_k(D) + \nu_k > \hat{T}\})$$

Let $g(D) = \max_{1 \leq i \leq k-1}\{f_i(D) + \nu_i\}$ then

$$\begin{aligned} P_{\hat{T},\nu_k}(A = a \mid V_1^{k-1}) &= P(\{\hat{T} \geq g(D)\} \bigcap \{f_k(D) + \nu_k > \hat{T}\}) \\ &= P(\{g(D) \leq \hat{T} \leq f_k(D) + \nu_k\}) \\ &= \int_{-\infty}^{\infty} P_{\hat{T}}(t) P_{\nu_k}(\nu) \mathbb{I}_{\{g(D) \leq t \leq f_k(D) + \nu\}} d\nu dt \end{aligned}$$

Let $\hat{t} = t - g(D) + g(D')$ then $g(D) \leq t \leq f_k(D) + \nu \longrightarrow g(D') \leq \hat{t} \leq f_k(D) + \nu_k + g(D') - g(D)$ and let $\hat{\nu} = \nu - f_k(D') + f_k(D) - g(D) + g(D')$ then $g(D) \leq t \leq f_k(D) + \nu \longrightarrow g(D') \leq \hat{t} \leq f_k(D') + \hat{\nu}$ and we have

$$\begin{aligned} P_{\hat{T},\nu_k}(A = a \mid V_1^{k-1}) = \iint P_{\hat{T}}\big(\hat{t} + g(D) - g(D')\big)\, P_{\nu_k}\big(\hat{\nu} + f_k(D') - f_k(D) + g(D) - g(D')\big) \\ \times\, \mathbb{I}\{g(D') \leq \hat{t} \leq f_k(D') + \hat{\nu}\}\, d\hat{\nu}\, d\hat{t} \end{aligned}$$

Recall that $\hat{T} = T + \text{Lap}(2/\varepsilon)$ and $\nu_k = \text{Lap}(4/\varepsilon)$ and that $g(D) = \max_{1\le i\le k-1}\{f_i(D) + \nu_i\}$ and by assumption the ℓ_1 sensetivity of f_i is 1 so $|f_i(D) - f_i(D')| \le 1$ and $|g(D) - g(D')| \le 1$ which means that

$$P_{\hat{T}}(t + g(D) - g(D')) \le e^{\frac{\varepsilon}{2}} P_{\hat{T}}(t)$$

and

$$P_{\nu_k}(\nu + f_k(D') - f_k(D) + g(D) - g(D')) \le e^{\frac{\varepsilon}{2}} P_{\nu_k}(\nu)$$

Therefore we have

$$\begin{aligned} P_{\hat{T},\nu_k}(A = a \mid V_1^{k-1}) &\le e^{\frac{\varepsilon}{2}} e^{\frac{\varepsilon}{2}} \iint P_{\hat{T}}(\hat{t}) P_{\nu_k}(\hat{\nu}) \mathbb{I}\{g(D') \le \hat{t} \le f_k(D') + \hat{\nu}\} d\hat{\nu} d\hat{t} \\ &= e^{\varepsilon} P_{\hat{T},\nu_k}(A' = a \mid V_1^{k-1}) \end{aligned}$$

Which implies that

$$P(A = a) \le e^{\varepsilon} P(A' = a)$$

and by symmetry we have

$$P(A' = a) \le e^{\varepsilon} P(A = a)$$

as desired. Since we just take expectations over $\nu_1, \ldots, \nu_{k-1}$. □

Accuracy of the Above Threshold Algorithm

Definition 3.24 (Accuracy for $\{\uparrow, \downarrow\}$ algorithms)**.** An algorithm with output stream $a_1, a_2, \ldots \in \{\uparrow, \downarrow\}$ in response to k queries is (α, β)-accurate if:

(1) The algorithm does not halt before k

(2) $\forall a_i = \uparrow$, $f_i(D) \ge T - \alpha$

(3) $\forall a_i = \downarrow$, $f_i(D) \le T + \alpha$

except with probability β. i.e

$$P(\{\text{doesnt halt before k}\} \cap \{\forall a_i = \uparrow\colon f_i(D) \ge T - \alpha\} \cap \{\forall a_i = \downarrow\colon f_i(D) \le T + \alpha\}) \ge 1 - \beta$$

Theorem 3.11. *For any sequence of queries $f_1, f_2, \ldots, f_k$ such that*

$$|\{i < k : f_i(D) \ge T - \alpha\}| = 0$$

AboveThreshold$(D, \{f_i\}, T, \varepsilon)$ is (α, β)-accurate with

$$\alpha = \frac{8(\ln(k) + \ln(2/\beta))}{\varepsilon}$$

Proof. We need the following:

(1) The first $(k-1)$ outputs should be $\downarrow$ i.e $f_i(D) + \nu_i \le \hat{T}$ i.e $f_i(D) \le T - \alpha \implies f_i(D) + \nu_i \le \hat{T}$

(2) If $f_i(D) + \nu_i \ge \hat{T}$ then $f_i(D) \ge T - \alpha$

(3) If $f_i(D) + \nu_i \le \hat{T}$ then $f_i(D) \le T + \alpha$

We show these:

(1) $f_i(D) \le T - \alpha$ so $f_i(D) + \nu_i \le T - \alpha + \nu_i$ I want $T - \alpha + \nu_i \le \hat{T} \implies \nu_i + T - \hat{T} \le \alpha$

(2) $f_i(D) + \nu_i \ge \hat{T} \implies f_i(D) \ge \hat{T} - \nu_i$ I want $\hat{T} - \nu_i \ge T - \alpha \implies \alpha \ge \nu_i + T - \hat{T}$

(3) $f_i(D) + \nu_i \leq \hat{T} \implies f_i(D) \leq \hat{T} - \nu_i$ I want $\hat{T} - \nu_i \leq T + \alpha \implies \alpha \geq \hat{T} - \nu_i - T$

So I want

$$\forall i \quad |\nu_i + T - \hat{T}| \leq \alpha$$

Equivalently,

$$\max_i |\nu_i + T - \hat{T}| \leq \alpha$$

Simplify to a sufficient condition,

$$\max_i |\nu_i| + |T - \hat{T}| \leq \alpha$$

so

$$\begin{aligned} &P(\max_i |\nu_i| + |T - \hat{T}| > \alpha) \\ &\leq P(\max_i |\nu_i| \geq \alpha/2 \cup |T - \hat{T}| \geq \alpha/2) \\ &\leq P(\max_i |\nu_i| \geq \alpha/2) + P(|T - \hat{T}| \geq \alpha/2) \end{aligned}$$

and $T - \hat{T} \sim \text{Lap}(2/\varepsilon)$ so

$$P(|T - \hat{T}| \geq \alpha/2) = P(|\text{Lap}(2/\varepsilon)| \geq \alpha/2) = e^{-\varepsilon \cdot \alpha/4}$$

and

$$P(\max_{1 \leq i \leq k} |\nu_i| \geq \alpha/2) = P(\exists i : |\nu_i| \geq \alpha/2) \leq \sum_{i=1}^{k} P(|\text{Lap}(4/\varepsilon)| \geq \alpha/2) = k \cdot e^{-\varepsilon \cdot \alpha/8}$$

Now plug in α so we get

$$P(\max_i |\nu_i| + |T - \hat{T}| > \alpha) \leq k \cdot e^{-\varepsilon \cdot \alpha/8} + e^{-\varepsilon \cdot \alpha/4} = k \cdot e^{-(\ln(k) + \ln(2/\beta))} + e^{-2(\ln(k) + \ln(2/\beta))} = \frac{\beta}{2} + \frac{\beta^2}{4k^2} < \beta$$

as desired. □

3.8.4 Summary of Composition

Remark 3.8. $k-$fold composition for $\varepsilon-$DP the mechanism is (ε', δ')-DP with

$$\varepsilon' = \varepsilon \sqrt{2k \ln\left(\frac{1}{\delta'}\right)} + k\varepsilon(e^{\varepsilon} - 1)$$

Corollary 3.1. *For $0 < \varepsilon' < 1$, to answer $k-$queries with budget (ε', δ') suffices to choose $\varepsilon = \frac{\varepsilon'}{2\sqrt{2k \ln(1/\delta')}}$*

Example 3.5. Say we want to answer $k-$queries with Laplace mechanism with sensitivity Δ. The naive composition we would use

$$f_i(D) + \text{Lap}(k \cdot \Delta/\varepsilon)$$

with standard deviation noise is $\Theta(k)$. Whereas if we use the advanced compositon theorem we would use

$$f_i(D) + \text{Lap}\left(\frac{2\Delta\sqrt{2k \ln(1/\delta')}}{\varepsilon}\right)$$

with standard deviation noise $\Theta(\sqrt{k})$.

3.9 Local Differential Privacy

Definition 3.25. Given $x \in \mathcal{X}$ and a trasformation $P_{Y|X}$ that gives output $Y \in \mathcal{Y}$ we say that $P_{Y|X}$ is ε-LDP if for all x, x' and all $S \subseteq \mathcal{Y}$,

$$P(Y \in S|X = x) \leq e^{\varepsilon} P(Y \in S|X = x') \iff D_{\infty}(P_{Y|X=x} \| P_{Y|X=x'}) \leq \varepsilon$$

Remark 3.9. We can visualize $P_{Y|X}$ as a matrix

$$P_{Y|X} = \begin{pmatrix} P(y_1|x_1) & \cdots & P(y_k|x_1) \\ \vdots & \ddots & \vdots \\ P(y_1|x_\ell) & \cdots & P(y_k|x_\ell) \end{pmatrix},$$

where row i is the output distribution given input x_i. The ε-LDP condition is equivalent to

$$P(y_j|x_i) \leq e^{\varepsilon} P(y_j|x_{i'}) \qquad \text{for all } i, i' \text{ and all } j,$$

that is, within each column of the matrix, the ratio of any two entries lies in $[e^{-\varepsilon}, e^{\varepsilon}]$. Fix $|\mathcal{X}| = \ell$ and $|\mathcal{Y}| = k$. Then

$$M_{\varepsilon} = \left\{ P_{Y|X} \in \mathbb{R}^{\ell \times k} : \begin{array}{ll} P_{Y|X}(y|x) \geq 0 & \forall x, y; \\ \sum_y P_{Y|X}(y|x) = 1 & \forall x; \\ P(y|x) \leq e^{\varepsilon} P(y|x') & \forall x, x', y \end{array} \right\}.$$

3.9.1 Histogram Estimation under LDP

Given $\mathcal{X} = \{1, \ldots, \ell\}$ we want to design $P_{Y|X}$ such that it is exactly ε-LDP. Some examples:

Example 3.6 (Warner's Randomized Response)**.** Let $\mathcal{X} = \{0, 1\}$ and $\mathcal{Y} = \{0, 1\}$. Then we can define $P_{Y|X}$ as follows:

$$P_{Y|X} = \begin{pmatrix} P(Y=0|X=0) & P(Y=1|X=0) \\ P(Y=0|X=1) & P(Y=1|X=1) \end{pmatrix} = \begin{pmatrix} \alpha & 1-\alpha \\ 1-\alpha & \alpha \end{pmatrix}$$

where $\alpha = \frac{e^{\varepsilon}}{e^{\varepsilon}+1}$.

Example 3.7 (l-ary Randomized Response)**.** Let $\mathcal{X} = \{1, \ldots, \ell\}$ and $\mathcal{Y} = \{1, \ldots, \ell\}$. Then we can define $P_{Y|X}$ as follows:

$$P_{Y|X} = \begin{pmatrix} P(Y=1|X=1) & P(Y=2|X=1) & \cdots & P(Y=\ell|X=1) \\ P(Y=1|X=2) & P(Y=2|X=2) & \cdots & P(Y=\ell|X=2) \\ \vdots & \vdots & \ddots & \vdots \\ P(Y=1|X=\ell) & P(Y=2|X=\ell) & \cdots & P(Y=\ell|X=\ell) \end{pmatrix} = \begin{pmatrix} p & q & q & \cdots & q \\ q & p & q & \cdots & q \\ q & q & p & \cdots & q \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ q & q & q & \cdots & p \end{pmatrix}$$

where $p = \frac{e^{\varepsilon}}{e^{\varepsilon}+(\ell-1)}$ and $q = \frac{1}{e^{\varepsilon}+(\ell-1)}$.

Definition 3.26 (One-hot Encoding)**.** For $\mathcal{X} = \{1, \ldots, \ell\}$ er define the encoding

$$enc : i \to e_i = (0, 0, \ldots, 1, 0, \ldots, 0) : \text{ where the 1 is in the ith position}$$

Algorithm 2 LDP Mechanism(x, ε)

Require: Input $x \in \{1, \dots, \ell\}$, privacy parameter ε
Ensure: $v(x) \in \{-1, 1\}^\ell$
Encode x as e_x
for $j = 1, \dots, \ell$ **do**
  if $j \neq x$ **then**
    $v_j(x) \sim \text{Unif}(\{-1, 1\})$
  else
    $v_x(x) \leftarrow \begin{cases} 1 & \text{with probability } \frac{e^\varepsilon}{e^\varepsilon+1} \\ -1 & \text{with probability } \frac{1}{e^\varepsilon+1} \end{cases}$
return $v(x)$

So the curator will observe $v^1, \dots v^n$ we define the histogram as follows:

$$f(x) = \sum_{i=1}^{n} \mathbb{I}_{\{X_i = x\}}, \quad f \in \mathbb{N}^\ell$$

and define the estimator

$$\hat{f}(v^1, \dots, v^n)(x) = \sum_{i=1}^{n} v^i(x) \cdot \frac{e^\varepsilon + 1}{e^\varepsilon - 1}$$

We need to check (1): this mechanism is ε-LDP and (2): $\mathbb{E}[\hat{f}(x)] = f(x)$.

Proof of (2).

$$\begin{aligned} \mathbb{E}[\hat{f}(x)] &= \sum_{i=1}^{n} \mathbb{E}[v^i(x)] \cdot \frac{e^\varepsilon + 1}{e^\varepsilon - 1} \\ &= \frac{e^\varepsilon + 1}{e^\varepsilon - 1} \cdot \sum_{i:X_i = x} \frac{e^\varepsilon + 1}{e^\varepsilon - 1} \\ &= \sum_{i=1}^{n} \mathbb{I}_{\{X_i = x\}} \\ &= f(x) \end{aligned}$$

□

Proof of (1). Fix $v \in \{-1, 1\}^\ell$ and $x \neq x'$ then

$$\begin{aligned} \frac{P(V = v | X = x)}{P(V = v | X = x')} &= \frac{\prod_{i=1}^{\ell} P(V_i = x_i | X = x)}{\prod_{i=1}^{\ell} P(V_i = x_i | X = x')} \\ &= \frac{P(V_x = v_x | X = x) \cdot P(V_{x'} = v_{x'} | X = x)}{P(V_{x'} = v_{x'} | X = x') \cdot P(V_x = v_x | X = x')} \\ &= \frac{P(V_x = v_x | X = x)}{P(V_x = v_x | X = x')} \\ &\leq \frac{\frac{e^\varepsilon}{e^\varepsilon+1}}{\frac{1}{e^\varepsilon+1}} \\ &= e^\varepsilon \end{aligned}$$

□

Chapter 4

Security

Motivation 4.1. The basic setup: Alice wants to send a message M to Bob over a noiseless channel, while an eavesdropper Eve sees everything that passes through it.

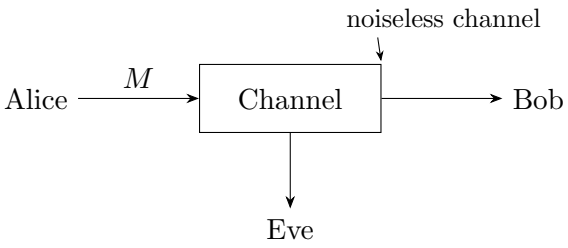


Figure 4.1: Standard security setup.

We want to keep M secret from Eve, so what can we do? There are three standard answers.

1. Shared secret key. Alice and Bob share a secret key K that Eve does not know. Alice encrypts M into a ciphertext C, and Bob uses K to recover M. Eve only sees C.

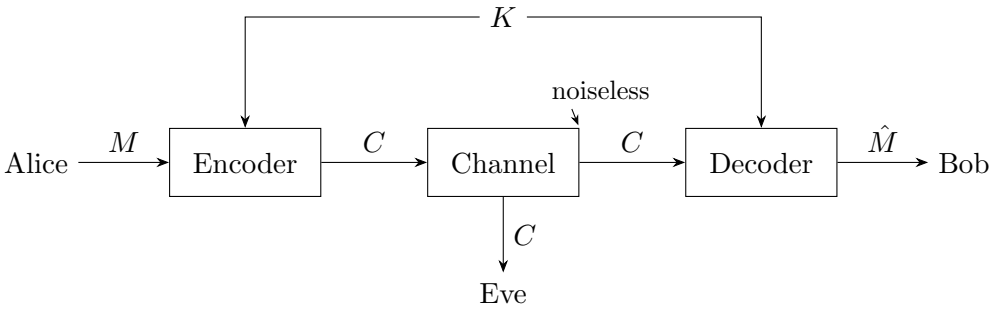


Figure 4.2: Symmetric-key setup: Alice and Bob share a secret key K.

2. Computational assumptions. If we assume Eve is computationally limited, Alice and Bob do not even need to share a key in advance: they can use public-key cryptography instead.

3. Noisy channels. Alternatively, we can use noise in the channel itself to keep M secret. This is the *wiretap channel*, the basic setup of information-theoretic security. Alice sends X over a noisy channel $P_{YZ|X}$; Bob receives Y, while Eve receives a different noisy version Z.

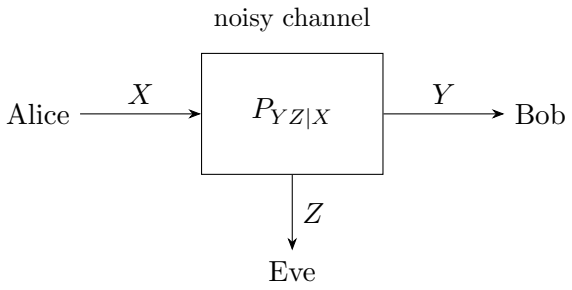


Figure 4.3: Wiretap channel.

4.1 Symmetric-key encryption

We start with the symmetric-key setup (Figure 4.2).

- The **message** M is a random variable on $\mathcal{M} = \{1, \dots, m\}$, with any distribution.
- The **key** K is uniform on $\mathcal{K} = \{1, \dots, k\}$ and independent of the message: $M \perp K$.
- Alice sends the **ciphertext**
$$C = \mathrm{Enc}(M, K),$$
and Bob decodes with
$$\hat{M} = \mathrm{Dec}(C, K).$$
- **Correctness:** Bob always recovers the message,
$$\mathrm{Dec}\big(\mathrm{Enc}(M, K), K\big) = M.$$

Perfect secrecy. The scheme has *perfect secrecy* if the ciphertext reveals nothing about the message:
$$C \perp M,$$

Example 4.1. Some examples are:

(1) $M = \{0, \dots, 25\}^n$ and $K \in \{0, \dots, 25\}$ we define
$$C = (M + K) \mod 26 = (M_1 + K, M_2 + K, \dots, M_n + K) \mod 26$$

(2) If $K \in \{0, 1, \dots, 26! - 1\}$ then K indicates a mapping
$$\{1, \dots, m\} \to \{1, \dots, m\}$$

(3) If $K \in \{0, \dots, 25\}^n$ then we can define
$$C = M \oplus K$$
Where $\oplus$ is sum modulo 26, which is known as the one-time pad or the vernam cipher. This achieves perfect secrecy.

Proposition 4.1. *If* $M \sim \mathrm{Unif}(\{0, \dots, m-1\}^n)$ *and* $K \sim \mathrm{Unif}(\{0, \dots, m-1\}^n)$ *and* $M \perp K$ *then*
$$C = M \oplus K \perp M$$

Proof. Fix $c \in \{0, 1, \dots, m-1\}$ and m_1, m_2 and consider

$$P_{M,C,K}(c \mid m_1) = P(m_1 \oplus K = c \mid m_1) = P(K = c \ominus m_1 \mid m_1) = P(K = c \ominus m_1) = \frac{1}{m^n}$$

Similarly,

$$P(c \mid m_2) = \frac{1}{m^n}$$

so

$$P(c \mid m_1) = P(c \mid m_2)$$

which implies that $C \perp M$ as desired. □

Theorem 4.1 (Shannon's Theorem)**.** *Let $M \sim \mathrm{Unif}(\mathcal{M})$ and $K \sim \mathrm{Unif}(\mathcal{K})$ with $K \perp M$, and suppose there exist* Enc *and* Dec *such that*

$$\mathrm{Dec}(\mathrm{Enc}(K, M), K) = M$$

and $C = \mathrm{Enc}(K, M)$ satisfies $C \perp M$ (perfect secrecy). Then $|\mathcal{K}| \geq |\mathcal{M}|$.

Proof. Let $\mathcal{C}$ be the set of possible ciphertexts. First assume that Enc is deterministic, and define

$$\mathcal{C}_m = \{c : \exists k \text{ such that } c = \mathrm{Enc}(k, m)\} \qquad \text{and} \qquad \mathcal{C}_k = \{c : \exists m \text{ such that } c = \mathrm{Enc}(k, m)\}.$$

To achieve perfect secrecy we must have

$$\mathcal{C}_{m_1} = \mathcal{C}_{m_2} = \mathcal{C} \quad \forall m_1, m_2 \in \mathcal{M},$$

since otherwise observing C would rule out some messages. Hence

$$|\mathcal{K}| \geq |\mathcal{C}_m| = |\mathcal{C}| \geq |\mathcal{C}_k|.$$

Now fix k and take $m_1 \neq m_2$. For decryption to succeed we need

$$\mathrm{Dec}(\mathrm{Enc}(k, m_1), k) = m_1 \neq m_2 = \mathrm{Dec}(\mathrm{Enc}(k, m_2), k),$$

so $\mathrm{Enc}(k, m_1) \neq \mathrm{Enc}(k, m_2)$. Thus $m \mapsto \mathrm{Enc}(k, m)$ is injective, so $|\mathcal{C}_k| = |\mathcal{M}|$, and therefore $|\mathcal{K}| \geq |\mathcal{M}|$.

Now suppose Enc is not deterministic, and define

$$\mathcal{C}_{k,m} = \{c : P(c \mid k, m) > 0\}, \qquad \mathcal{C}_k = \bigcup_m \mathcal{C}_{k,m}, \qquad \mathcal{C}_m = \bigcup_k \mathcal{C}_{k,m}.$$

As before, perfect secrecy gives $\mathcal{C}_m = \mathcal{C}$ for all m, and decodability gives $\mathcal{C}_{k,m_1} \cap \mathcal{C}_{k,m_2} = \emptyset$ for $m_1 \neq m_2$. So

$$\begin{aligned}
|\mathcal{C}| &\geq \max_k |\mathcal{C}_k| = \max_k \Big|\bigcup_m \mathcal{C}_{k,m}\Big| = \max_k \sum_m |\mathcal{C}_{k,m}| \geq \frac{1}{|\mathcal{K}|} \sum_{m,k} |\mathcal{C}_{k,m}| \\
&= \frac{|\mathcal{M}|}{|\mathcal{K}|} \cdot \frac{1}{|\mathcal{M}|} \sum_m \sum_k |\mathcal{C}_{k,m}| \geq \frac{|\mathcal{M}|}{|\mathcal{K}|} \min_m \sum_k |\mathcal{C}_{k,m}| \\
&\geq \frac{|\mathcal{M}|}{|\mathcal{K}|} \min_m \Big|\bigcup_k \mathcal{C}_{k,m}\Big| = \frac{|\mathcal{M}|}{|\mathcal{K}|} \min_m |\mathcal{C}_m| = \frac{|\mathcal{M}|}{|\mathcal{K}|} |\mathcal{C}|.
\end{aligned}$$

Dividing by $|\mathcal{C}|$ gives $|\mathcal{K}| \geq |\mathcal{M}|$. □

Remark 4.1. We want to relax the notation of perfect secrecy into parial secrecy. We also need a measure of information between M and C.

4.2 Information Theory

4.2.1 Entropy

Definition 4.1 (Entropy)**.** For a discrete random variable $X \in \mathcal{X}$ with distribution P_X we define entropy as

$$H(X) = \sum_x P(x) \log \frac{1}{P(x)} = \mathbb{E}_{P_X}\left[\log \frac{1}{P_X(X)}\right]$$

Example 4.2. IF $X \sim \text{Ber}(p)$ then $H(X) = p \log \frac{1}{p} + (1-p) \log \frac{1}{1-p} = -p \log p - (1-p) \log(1-p)$ Sometimes called the binary entropy function.

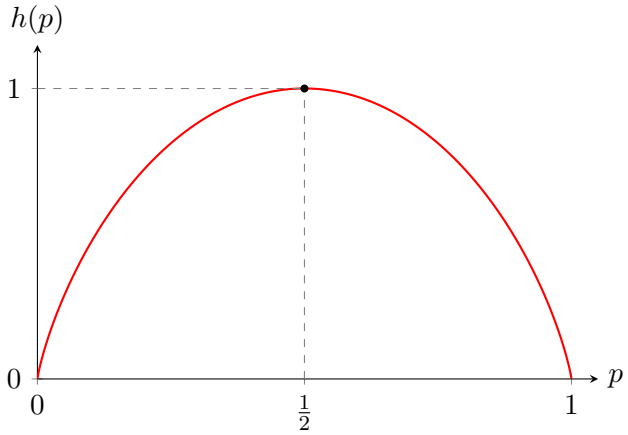


Figure 4.4: The binary entropy function $h(p)$.

Example 4.3. Given $X = \{0, 1, 2, 3\}$ with $P_X = [0.1, 0.2, 0.3, 0.4]$ then

$$H(X) = 0.1 \log 10 + 0.2 \log 5 + 0.3 \log \frac{10}{3} + 0.4 \log \frac{5}{2} \approx 1.846$$

Properties 4.1 (Properties of Entropy)**.** Some important properties of entropy are:

(1) $H(X) \geq 0$ with equality if and only if X is deterministic.

(2) $H(X) \leq \log |\mathcal{X}|$ with equality if and only if $X \sim \text{Unif}(\mathcal{X})$.

Proof. We prove each property in turn.

(1) Since $P(x) \in [0, 1]$, we have $\log \frac{1}{P(x)} \geq 0$ whenever $P(x) > 0$ (with the convention $0 \log \frac{1}{0} = 0$). Hence every term of

$$H(X) = \sum_x P(x) \log \frac{1}{P(x)}$$

is non-negative, so $H(X) \geq 0$.

If X is deterministic, then $P(x_0) = 1$ for some x_0 and $P(x) = 0$ for all $x \neq x_0$, so

$$H(X) = 1 \cdot \log 1 + 0 = 0.$$

Conversely, suppose $H(X) = 0$. Since $H(X)$ is a sum of non-negative terms, each term must vanish: for every x, either $P(x) = 0$ or $\log \frac{1}{P(x)} = 0$, i.e. $P(x) \in \{0, 1\}$. As $\sum_x P(x) = 1$, exactly one x_0 has $P(x_0) = 1$, so X is deterministic.

(2) By non-negativity of relative entropy,

$$\begin{aligned} 0 \le D\big(P_X \parallel \mathrm{Unif}(\mathcal{X})\big) &= \sum_x P(x) \log \frac{P(x)}{1/|\mathcal{X}|} \\ &= \sum_x P(x) \log P(x) + \sum_x P(x) \log |\mathcal{X}| \\ &= -H(X) + \log |\mathcal{X}|. \end{aligned}$$

Hence $H(X) \le \log |\mathcal{X}|$, with equality if and only if $D\big(P_X \parallel \mathrm{Unif}(\mathcal{X})\big) = 0$, which holds if and only if $P_X = \mathrm{Unif}(\mathcal{X})$.

□

4.2.2 Operational interpretation of entropy: Compression

Theorem 4.2. *If $X_1, \ldots, X_n$ are i.i.d. with distribution P, then we can look at*

$$-\frac{1}{n} \log P(X_1, \ldots, X_n) \to H(X)$$

More precisely, for all $\varepsilon > 0$

$$\lim_{n\to\infty} P_{p^{\otimes n}} \left(\left| -\frac{1}{n} \log P(X_1, \ldots, X_n) - H(X) \right| > \varepsilon \right) = 0$$

Proof.

$$-\frac{1}{n} \log(P(X_1, \ldots, X_n)) = \frac{1}{n} \sum_{i=1}^{n} -\log P(X_i)$$

Let $\sigma^2 = \mathrm{Var}(P(X))$ then by WLLN we have

$$\frac{1}{n} \sum_{i=1}^{n} -\log P(X_i) \to \mathbb{E}[-\log P(X)] = H(X)$$

in probability as $n \to \infty$. □

Definition 4.2. A typical set $A_\varepsilon^{(n)}$ with respect to P is defined as

$$A_\varepsilon^{(n)} = \{x^n : 2^{-n(H(X)+\varepsilon)} \le P(x^n) \le 2^{-n(H(X)-\varepsilon)}\}$$

for some $\varepsilon > 0$.

Properties 4.2. Some important properties of typical sets are:

(1) $x^n \in A_\varepsilon^{(n)}$ then $H(x) - \varepsilon \le -\frac{1}{n} \log P(x^n) \le H(x) + \varepsilon$

(2) $P(A_\varepsilon^{(n)}) \ge 1 - \varepsilon$ for sufficiently large n.

(3) $|A_\varepsilon^{(n)}| \le 2^{n(H(X)+\varepsilon)}$ for sufficiently large n.

(4) $|A_\varepsilon^{(n)}| \ge (1-\varepsilon) 2^{n(H(X)-\varepsilon)}$ for sufficiently large n.

Proof. We prove each property in turn.

(1) This follows directly from the definition of $A_\varepsilon^{(n)}$: for $x^n \in A_\varepsilon^{(n)}$,

$$2^{-n(H(X)+\varepsilon)} \le P(x^n) \le 2^{-n(H(X)-\varepsilon)},$$

which is equivalent to $H(X) - \varepsilon \le -\frac{1}{n}\log P(x^n) \le H(X) + \varepsilon$.

(2) Since $X_1, \dots, X_n$ are i.i.d.,

$$-\frac{1}{n}\log P(X^n) = \frac{1}{n}\sum_{i=1}^{n} -\log P(X_i),$$

which is an average of i.i.d. random variables with mean $\mathbb{E}[-\log P(X)] = H(X)$. By the weak law of large numbers,

$$\Pr\left(\left|-\frac{1}{n}\log P(X^n) - H(X)\right| \le \varepsilon\right) \to 1,$$

so $P\big(A_\varepsilon^{(n)}\big) > 1 - \varepsilon$ for all sufficiently large n.

(3) Using the lower bound in (1),

$$1 = \sum_{x^n \in \mathcal{X}^n} P(x^n) \ge \sum_{x^n \in A_\varepsilon^{(n)}} P(x^n) \ge \big|A_\varepsilon^{(n)}\big|\, 2^{-n(H(X)+\varepsilon)},$$

so $\big|A_\varepsilon^{(n)}\big| \le 2^{n(H(X)+\varepsilon)}$.

(4) Using (2) and the upper bound in (1), for n sufficiently large,

$$1 - \varepsilon < P\big(A_\varepsilon^{(n)}\big) = \sum_{x^n \in A_\varepsilon^{(n)}} P(x^n) \le \big|A_\varepsilon^{(n)}\big|\, 2^{-n(H(X)-\varepsilon)},$$

so $\big|A_\varepsilon^{(n)}\big| \ge (1-\varepsilon)\, 2^{n(H(X)-\varepsilon)}$.

□

Summary 4.1. We have the following properties of typical sets:

(1) $|A_\varepsilon^n| \approx 2^{nH(X)}$ for sufficiently large n.

(2) $P(A_\varepsilon^n) \to 1$

(3) all sequences consentrate in a relatively small set of sequences (P_X is not uniform)

(4) Inside the typical set, all sequences have roughly the same probability.

Definition 4.3 (Compression Scheme)**.** A *compression scheme* is a map

$$\mathrm{Enc} : \mathcal{X}^n \to \{0,1\}^*.$$

It is *valid* if there exists Dec such that

$$\mathrm{Dec}\big(\mathrm{Enc}(x^n)\big) = x^n \qquad \text{for all } x^n \in \mathcal{X}^n.$$

Proposed scheme. Let $\ell(x^n)$ denote the length of $\mathrm{Enc}(x^n)$, where

$$\ell(x^n) = \begin{cases} 1 + \left\lceil \log\left|A_\varepsilon^{(n)}\right| \right\rceil \;\le\; 2 + n\big(H(X)+\varepsilon\big) & \text{if } x^n \in A_\varepsilon^{(n)}, \\ 1 + \lceil n \log |\mathcal{X}| \rceil & \text{otherwise.} \end{cases}$$

Then

$$\frac{1}{n}\,\mathbb{E}_{P_{X^n}}\big[\ell(X^n)\big] \;\le\; H(X) + \varepsilon + \frac{2}{n} + \varepsilon\Big(\log|\mathcal{X}| + \frac{2}{n}\Big).$$

As $n \to \infty$,

$$\frac{1}{n}\,\mathbb{E}_{P_{X^n}}\big[\ell(X^n)\big] \;\le\; H(X) + \varepsilon\big(1 + \log|\mathcal{X}|\big).$$

Since ε was arbitrary, letting $\varepsilon \to 0$ gives

$$\frac{1}{n}\,\mathbb{E}_{P_{X^n}}\big[\ell(X^n)\big] \;\le\; H(X),$$

and therefore

$$\lim_{n\to\infty} \inf_{\mathrm{Enc}:\mathcal{X}^n \to \{0,1\}^*} \frac{1}{n}\,\mathbb{E}\big[\ell(X^n)\big] \;\le\; H(X).$$

Proposition 4.2. *Let $X_1, \ldots, X_n$ be i.i.d.$\sim P_X$. Let*

$$B_\delta^{(n)} = \min\{|B| : B \subseteq \mathcal{X}^n \text{ and } P(B) \ge 1-\delta\}$$

Then

$$\lim_{n\to\infty} \frac{1}{n} \log B_\delta^{(n)} \ge H(X)$$

4.2.3 Joint Entropy

Definition 4.4. Given a pair of random variables (X, Y) with joint distribution P_{XY} we define the joint entropy as

$$H(X,Y) = -\sum_{(x,y)} P(x,y) \log P(x,y)$$

Proposition 4.3. *Special Cases:*

(1) If $X \perp Y$ then $H(X,Y) = H(X) + H(Y)$

(2) $H(X,X) = H(X)$

Proof. (1) If $X \perp Y$, then $P(x,y) = P(x)P(y)$ for all x, y. Thus,

$$\begin{aligned} H(X,Y) &= -\sum_{(x,y)} P(x,y) \log P(x,y) \\ &= -\sum_{(x,y)} P(x)P(y) \log\big(P(x)P(y)\big) \\ &= -\sum_{(x,y)} P(x)P(y)\big(\log P(x) + \log P(y)\big) \\ &= -\sum_x P(x) \log P(x) - \sum_y P(y) \log P(y) \\ &= H(X) + H(Y). \end{aligned}$$

(2) For the second case, we have $P(x,x) = P(x)$ and $P(x,y) = 0$ for $x \neq y$. Thus,

$$\begin{aligned} H(X,X) &= -\sum_{(x,x')} P(x,x') \log P(x,x') \\ &= -\sum_{x} P(x)P(x|x) \log P(x)P(x|x) \\ &= -\sum_{x} P(x) \log P(x) \\ &= H(X). \end{aligned}$$

□